A Cheap Model and a Deterministic Checker: Fewer Silent Errors in Numeric Document Extraction from the Documents' Own Identities

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Abstract. Investors, customers and limited partners of asset managers, private equity and venture capital firms, and alternative market data providers receive a steady stream of financial statements, from quarterly packets for each private fund to the 10-Qs and 10-Ks of public companies, and cheap vision-language models are increasingly used to extract the numbers. Those numbers are usually right; the costly ones are wrong and unflagged, which we call *silent*. Many document sets constrain their own numbers through linear identities $Ax = b$: totals, roll-forwards, quantity times price, the same figure printed twice. We put a deterministic checker between a cheap model and a stronger model or a person. It changes a value only when an identity, or an unambiguous and corroborated read of the printed page, backs the change, and it sends every other doubt on with its evidence. Nothing in it names a domain, and it can discover its identities from the page with no configuration. Each version is tuned on development data, frozen and run once on held-out data. The latest, qcore9 (first freeze, sha256 `3ca510d3...`), was run once on 24 SEC filings locked before its freeze (72 statements, 5,351 numbers, 16 10-Qs and 8 10-Ks), read from poor scans that keep a born-digital text layer. There Claude Haiku with the checker leaves 9 silent errors (95% CI 0–26) where Claude Opus alone leaves 25; on this set alone the paired difference, -16 (-43 to $+10$), is not significant, because Opus made few errors. On the 22 filings of the previous real test it was 15 against 50 (-35, -73 to -6), and pooled over four held-out SEC sets (227 statements, 17,125 numbers) it is 28 against 174 (-146, -231 to -72; 32 if the earlier version is used on all four). On the new filings Opus with the checker leaves no silent error and GPT goes from 1,090 to 15. In the first real held-out test of the final zero-config mode, with no identity or schema file, Haiku leaves 16 silent errors and GPT 37 (qcore8: 51), more than with hand-written identities (9 and 15); every zero-config run with the text layer broke the same three numbers, cash-flow totals whose closing parenthesis the filing's text layer prints as a separate word. On synthetic held-out sets zero-config left 2 to 7 times the silent errors of hand identities. A re-read of unsure cells by a vision model is not independent evidence: its errors are correlated with the extraction's; the identities' are not. Without a trustworthy text layer the cheap model does not overtake the strong one.

1 Introduction

Investors, customers and limited partners of asset managers, private equity and venture capital firms, and alternative market data providers receive a steady stream of financial statements: a quarterly packet of statements from each private fund, and the quarterly and annual filings of every public company they follow. Cheap vision-language models are increasingly used to extract the numbers, and the numbers feed valuations, investor reports and trades. These documents, like many others, are internally redundant. A balance sheet prints its totals; a cash-flow statement rolls a balance forward; an invoice prints line amounts, a subtotal, tax and a total; a receipt prints quantity, unit price and amount; a statistical bulletin prints rows and the column that adds them. Write the extracted values as a vector x: the document set asserts $Ax = b$ up to rounding. This paper is about using that redundancy, and the printed page, to catch the errors a model makes without telling anyone.

On clean pages the strongest models rarely err. On 21,367 numbers in clean SEC notes tables Claude Opus made no error, and on 7,892 numbers of synthetic reports, invoices and bulletins it made none either (Section 5.7). Reading is not always easy: a pipeline rasterises a filing at low resolution, a statement arrives as a fax, a receipt as a phone photo. On poor scans of 227 held-out SEC statements Opus is 98.98% accurate, Claude Haiku 95.94% and GPT 76.20%. A wrong number that some check points at costs a reviewer a minute; one that nothing points at goes into a model, a filing or a payment. We count those *silent* errors. Strong models' errors can be harder to catch, partly because the models say that when a digit is unclear they pick the value that makes the totals add up (Section 5.9). Any check that only sums the extraction accepts that number.

We describe a ladder (Figure 1). A cheap model extracts. A deterministic checker, which uses no model, reads the page, tests the identities, and writes a value only when an identity backs the change or the read of that exact cell is unambiguous and corroborated; it may write only a number the field's own line prints or one already extracted. Everything else goes up as a small packet with its evidence, to a stronger model or a person. The method needs no domain code: the identities and field labels are inputs, and the sixth to ninth versions discover them from the documents themselves when none are given. Our findings:

1. **Replication on new filings, twice.** On 22 filings built after qcore6 was frozen, Haiku plus the checker leaves 15 silent errors against Opus alone's 50 (-35, -73 to -6). On 24 further filings locked before qcore9 was frozen it leaves 9 against 25: fewer, but not significantly (-16, -43 to $+10$), as Opus alone erred little there. Pooled over four SEC sets it is 28 against 174 (-146, -231 to -72) (Section 5.1).
2. **It helps every model.** Opus, Haiku and GPT, eight held-

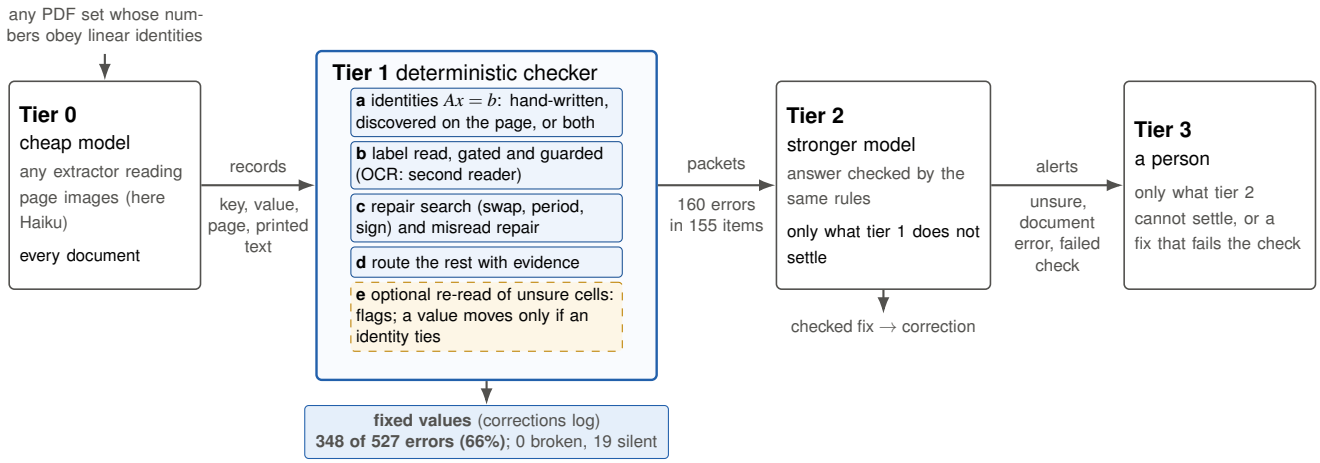


Figure 1: The tier ladder. A cheap model extracts; a deterministic checker writes a value only when the page and the documents' own linear identities agree, and sends everything else on as small packets with evidence; a stronger model or a person sees only those. Nothing in tier 1 names a domain. Counts: Claude Haiku on the 155 held-out SEC statements read from poor scans (11,774 numbers; frozen qcore6, hand identities, born-digital text layer).

out sets: significant cuts in 50 of 60 per-set settings with qcore6 and 11 of 12 with qcore9 on the newest filings, where Opus's silent errors fall from 25 to 0 and GPT's from 1,090 to 15. Every broken number is listed with its cause (Section 5.2).

3. **Zero configuration works, but less well.** On the newest filings, with no identity or schema file, the checker leaves Haiku 16 silent errors and GPT 37, against 9 and 15 with hand-written identities and 168 and 1,090 for the models alone; it breaks three correct numbers per model where a text layer misprints signs. On synthetic held-out sets it leaves 2 to 7 times the silent errors of hand identities, mostly on layouts it was never tuned on (Sections 5.3 and 5.4).
4. **A second read is not independent evidence.** A vision-model re-read of unsure cells repeats the extraction's wrong digit often enough (32% of wrong cells for a Haiku reader, 16% for Opus, on development data) that we let it move a value only through an identity. Held out, it changed nothing and made a few errors visible (Section 5.5).
5. **Where it stops.** Without a born-digital text layer the cheap model does not catch up, on any SEC set, though the checker still cuts the strong model's silent errors (Section 5.6).

We claim no novelty for the repair operations (Section 2). What we add is the page-grounded gate, identity discovery validated on the page, a measured account of model-on-model re-reading, and an evaluation protocol that freezes code before every held-out run.

2 Related work

Correcting sign errors, swapped values and typing errors under linear balance edits is established in official statistics and is the direct prior art for our repair step. Scholtus finds a minimal set of sign flips and swaps that makes a record consistent, and accepts a rounding repair only if it creates no new edit violations [24]; an earlier report corrects single-digit typing errors [23]. The deducorrect package implements typo, sign and swap correction and touches no variable in a satisfied edit [27]; its acceptance rule is ours. The deductive, automatic, then manual editing process [21] has the shape of our tiers, and error localisation goes back to Fellegi and Holt [10]. DART repaired OCR'd balance sheets under aggregate constraints with an operator accepting each update [5, 9, 11]; an IPUMS tool locates OCR digit errors in census tables from row and column totals [19]; Correia and Luck flag OCR'd balance sheets with accounting identities [7]. These impute any value that satisfies the constraints, or only flag; ours admits only values the page prints on the field's own line or that were already extracted.

Hemmer et al. filter LLM extractions of CORD and SROIE receipts with arithmetic constraints [15], and Aggarwal et al. flag failing sums in fiscal documents [2]. Financial verification systems check claims or detect inconsistencies in statements and XBRL [13, 20, 26, 28, 29]; none repairs extracted values against the page. Totals in early XBRL filings often did not equal their components [8], one reason not to take filed identities on trust. We use CORD [22], Donut [18] and the VRDU ad-buy forms [30]. VLMs invent characters on degraded documents [14], and OCR noise corrupts numbers in LLM key-value extraction [3]. Model cascades route on learned scores, self-verification or confidence [1, 6, 12, 17, 25]; our gate is a deterministic identity test plus page evidence. LLMs do not reliably self-correct without external feedback [16], and models satisfy a visible check rather than the intended goal [4, 31]; our re-read results add that a second model read of the same pixels is correlated with the first.

3 Method

3.1 Inputs and identities

A *bundle* is a set of PDFs, optionally with a schema and a list of identities (truth is used only for scoring). The schema

gives each field's printed labels, units and sign convention. A record's key is its document, field, period, entity (a row-group heading such as a segment or a receipt item) and column. An extraction is a list of key, value, page and printed text; a missing key is a blank. Domain knowledge lives only in the schema, the identities and a small adapter per data source; the core names no field, document or dataset.

Identity k has a coefficient row a_k on inputs I_k, a constant b_k and a tolerance τ_k (half a printed rounding unit per term):

$$g_k(x) = a_k^\top x - b_k, \qquad k \text{ ties} \iff |g_k(x)| \le \tau_k. \tag{1}$$

Product terms (count times unit price) are allowed; search and repair then use the partial derivative as the coefficient. An identity with a blank input is skipped, not failed. $\mathscr{F}(x)$ and $\mathscr{T}(x)$ are the failing and tying identities. In our data the identities are a 10-Q filer's XBRL calculation linkbase plus the balance-sheet equation and cash roll-forward; receipt arithmetic; bulletin row and column sums; a fund's roll-forwards and accumulation across quarters; or, in zero-config mode, whatever Section 3.5 finds.

3.2 Label read, gate and guard

Before any arithmetic the checker reads the page. For each record it finds the field's *own line*, the line whose words best match a label, and the column from the header cell above; if no number or two numbers sit under that header, the read is empty. A unit statement ("in thousands") sets a scale s, and a printed p matches v when $|p\,s - v| \le s/2$. If the printed $p_i \ne x_i$, the read proposes $x_i \leftarrow p_i$.

A proposal passes the *gate* only under one of two rules. (a) *Identity-backed*: it makes an identity through the cell tie that did not, and breaks none that tied. (b) *Unambiguous and corroborated*: the label matches exactly one line in the document, the header places the cell, no other column lands on it, and the extraction corroborates the read (a sign flip, a one-glyph misread, a blank filled, or the extracted number printed elsewhere on the row or nowhere in the document). A read never changes a value by a power of 1000. Passing reads are applied jointly under a *guard*: a tied identity may break only if every read feeding it is backed by its field's single best line. Every other read becomes a tier 2 item with its reason.

3.3 Repair search and routing

For each failing identity and input i, the search tries swapping x_i with another number of the document, with the same line in the adjacent period, and negating it. A candidate x' counts only if

$$\mathscr{F}(x') \subsetneq \mathscr{F}(x) \tag{2}$$

and the page does not contradict it. A closing value $v = x_i - g_k(x)/a_{ki}$ is kept only if every failing identity through i implies it, it is a digit-level misread of x_i, (2) holds and v is printed on i's own line. So every value the checker writes lies in

$$x_i' \in \{p_i\} \cup \{x_j, -x_j : j \text{ extracted}\} : \tag{3}$$

Checker (one document; extraction x, identities $Ax = b$, page reads p)

1: Label each read: *corroborated* if x_i matches the page on i's own line.
2: $\mathscr{F}(x) \leftarrow \{k : |g_k(x)| > \tau_k\}$, where $g_k(x) = a_k^\top x - b_k$.
3: **for** each $k \in \mathscr{F}(x)$ and input i of k **do**
4: candidates x': swap x_i with another number, with the adjacent period's value, or negate it; or the closing value $v = x_i - g_k(x)/a_{ki}$.
5: **keep** x' only if $\mathscr{F}(x') \subsetneq \mathscr{F}(x)$ (2), the page does not contradict it, and x_i' lies in the allowed set (3).
6: **gate:** write x' only if (a) it is backed by an identity or (b) the read is unambiguous and corroborated.
7: **for** each k still in $\mathscr{F}$ **do**
8: **if** every input of k matches the page: mark a source-document error; keep values.
9: **else** route to tier 2, naming only inputs not shown right.
10: **return** x, tier-2 items.

Figure 2: The checker's decision steps. It never writes a value outside the allowed set, so it cannot invent a number to make the books tie.

it never invents a number to make the books tie. Remaining failures become tier 2 items naming only the inputs not shown to be right; when every input matches the page, the item is a source-document error and the values stay. Without page text the checker writes nothing except a flip back to a declared sign. Figure 2 summarises the steps.

3.4 What the checker reads

Text mode reads the PDF's text layer while the model reads a page image: a born-digital PDF that a pipeline rasterises (for CORD, the dataset's OCR annotation drawn as invisible text). *No-text mode* runs the identities alone. *OCR mode* is for scans: the OCR (docTR or Tesseract; for the real scans, the dataset's OCR or the scanner's layer) is a second, noisy reader. Where it agrees with the model the cell is never changed; where they disagree, the candidates are the OCR read, its one-glyph look-alikes and digit-wise mixes, and a change set is taken only if it is the unique best one under the identities. New in qcore6: a PDF whose page is covered by an image is a scan, and its text layer is treated as an OCR, not as the page; a read there is corroborated only by its digits. On development scans qcore5 broke 4 correct numbers per model in text mode this way and qcore6 breaks none.

3.5 Zero-config: identities and schema discovered on the page

With `-identities auto` and `-schema auto` the bundle holds only the PDFs. Each document's records are laid out as a grid in page reading order, and candidate identities are generated as templates: *row sums* (subtotals, totals and roll-forwards: spans of up to 25 rows above a row, every sign pattern up to four terms, collapsed so that a total over subtotals is found after them), *column sums* (segments adding to a

consolidated column), *products* (quantity times price equals amount) and *equal ties* across tables and documents (the same number at least 100 rounding units, same period, same duration and years in the headings, a shared label word). A template is checked on every instance, reading each cell from the page where it can and from the extraction otherwise; it is admitted if it holds on at least two instances and fails on at most $\max(1, 10\%)$, and refused if the page reads it whole somewhere and never confirms it, so an identity is never invented from the extraction where the page disagrees. Field labels come from the extraction's printed text and are kept only if a page line with every word of the label prints the field's number; units and sign conventions are inferred the same way. On development data 99.7% of the identities discovered in text mode held on the true values (12,401 of 12,437) and 99.1% without (7,093 of 7,154). `-identities both` adds the discovered identities to a hand-written file.

qcore7 and qcore8. Both change only what runs when something is discovered (`auto` or `both` identities, or `auto` schema) and add no new way to change a value. qcore7 keeps label cells from being read as values (age bands, years, numbers inside a label), requires a tier 1 change resting only on discovered identities to have independent backing (a hand identity, two discovered identities that do not share most of their inputs, or one confirmed on the page in two places), adds aggregate ties, and lets the inferred schema accept *weak* labels that change a value only through an identity or as a one- or two-glyph misread. qcore8 admits more templates: a sum of four or more terms is admitted when it holds with large numbers in at least two places and fails no more often than it holds, and the page must confirm it four times as often as it refutes it where it reads every cell; it adds transposed templates, roll-forwards over the previous column, sparse sums, closing balances carried to the next statement, year-to-date accumulation and products with a rate; it sets a tolerance from the finest rounding unit two numbers show; and a weak label changes a value only through an identity. Details: Appendix J and Supplement S10.

qcore9, the version tested on set 4. Five generic changes, made on development data (Appendix J, Supplement S10); again no new way to change a value. (1) A sign proposal from the label read is dropped when the extraction's own citation is another print of the same number showing the extracted sign (a bank statement prints each check as (542.03) in one list and 542.03 in another). (2) When a candidate line spells every identifier of a record in its label ('ANA-128'), the other candidate lines are dropped. (3) A printed number matches a stored value at the page's scale only within half of the printed number's own last digit, not within half a page unit, and a percentage never does: '0.4' on a page in thousands had matched any value from -100 to 900. (4) A record with an entity owns only the lines that name the entity. (5) Without page text, discovery admits a plain sum whose every failure is one slip of one cell, checks long sums of large numbers that foot exactly elsewhere to 1.5 units, tries a total as the sum of the section totals above it and admits equal ties between non-adjacent totals; and a page read that rule (b) takes on its own keeps its page basis when the only identity through it is discovered. Changes 1, 2, 4 and 5 run only in zero-config code. Change 3 runs in every mode that reads the page, so with hand identities qcore9, unlike qcore7 and qcore8, does not always return qcore6's values (Section 5.1).

3.6 Re-read tier and its change gate

For the cells of tier 2 items (failing identities, misreads, reader disagreements, identities that tie while the OCR disagrees on a cell, blanks; at most 0.6 requests per number) the checker cuts a crop: the column header strip stitched above the cell's row, the cell outlined, deskewed and scaled. A row check moves or drops crops whose row does not print the record's label. One question asks a vision model for the printed string, the row label and the column heading; an answer naming another row or year is no evidence. The answer is a third reader of that cell, and the gate stays: a re-read may move a value only when it agrees with the page read of that exact cell *and an identity through the cell then ties* (rule a). A re-read agreeing with the extraction clears a flag only when the page read is a different number altogether, not a glyph or two away. A disagreeing re-read becomes a tier 2 item with every read attached. We set this gate on development data after real reads showed why (Section 5.5).

4 Evaluation protocol and data

4.1 Freeze, tune on development data, test once

Every version was frozen (sha256 of every file in a manifest), tuned on development data with generic fixes only, frozen again and run once on held-out data; problems found there are reported, not fixed (Table 1). qcore6 (28 files, combined sha256 `86debea05f95a6d2d93a7add48d32cd54e6109135052df4c1dd178c48a95ab34`, frozen 2026-10-16T13:14Z, 56 tests) was tuned on development splits only; its change log lists the held-out data it did not open. The hash matched before the runs, at their end and at report time; 442 runs (qcore6 338, frozen qcore5 re-run 104) each ran once, with no crash. With the hand-written identities qcore6 returned exactly qcore5's fixes and tier 2 items in all 104 (set, model, mode) runs, so the hand-identity results below are also a test of qcore5's frozen rule set; qcore6's new held-out evidence is the discovered identities and the re-read tier. SEC set 1 was seen before qcore4 and qcore5 were designed; SEC sets 2 and 3, the notes tables, the real scans and the synthetic bundles 02–05 are clean tests, and set 3 was built after qcore5 was frozen.

qcore7 (28 files, sha256 `51d9a352...`, frozen 2026-10-16T16:40Z, 65 tests) and qcore8 (29 files, `40518616...`, 21:54Z, 71 tests) were tuned on every set above, all spent by then, and each was run once on a fresh synthetic set locked by a hash manifest before any of its changes (432 and 1,296 runs, no crash). qcore8 was frozen three times before its run, each time after a development break. With hand identities qcore7

and qcore8 returned exactly qcore6's fixes and tier 2 items in every held-out and development run.

qcore9 (30 files, combined sha256 `3ca510d3ea178550ee3a43676d9f8e74a4e4ec0c4e4e8b7d4149d72732d387e0`, frozen 2026-10-18T18:27Z, 79 tests) was tuned on every set above, qcore8's fresh set included. This first freeze is the version run on set 4 and the one every qcore9 result here describes. Its held-out test is SEC set 4. The set was built and locked (bundles 15:44Z) and its Haiku and Opus extractions (15:54Z) and GPT extractions (16:12Z) were locked by hash manifests before the freeze; qcore9's change log records that neither the filings nor the set were opened while it was written. Frozen qcore9, qcore8 and qcore6 were run once on it (90 runs, 18:40–19:19Z, no crash, none repeated), with the three package hashes and the four lock manifests verified before and after. The scripts were written before any result was seen, and the modes, identity settings and scoring are set 3's. The re-read tier was not run on set 4: it needs an Opus reader answering crop requests, which that session could not run. *A later refreeze.* The first freeze's development grid, run alongside, broke one new number (CAT's net income to common holders, 6,142 to 6,141 M, in discovered-identity text runs on SEC development filings), and qcore9 was refrozen at 19:59Z (freeze 2, `c4401b94...`) with one change in discovery (a one-unit difference no longer counts as a one-digit slip). Its freeze record still lists both freezes as before any held-out run, but by the time stamps the second came 40 minutes after the set-4 run had ended (19:19Z); a note added to the record says freeze 2 was made from the first freeze's development grid only, without seeing held-out results. Freeze 2 has not been run on set 4 or any held-out data; every qcore9 result here is freeze 1's. Its code is kept as `qcore9_f1/` (package name `qcore9`, since `qcore9/` now holds freeze 2), rebuilt by reverting the post-freeze edits; copied to a folder named `qcore9` it hashes to `3ca510d3...` exactly and its 79 tests pass.

4.2 Data and models

SEC statements, poor scans. Balance sheet, income statement and cash-flow statement of held-out filings: *set 1*, 10 filers' 10-Qs (29 statements, 2,144 numbers); *set 2*, 20 other filers' 10-Qs (60, 4,424); *set 3*, 22 further filers across utilities, energy, materials, industrials, consumer, health care and financials, 15 10-Qs and 7 10-Ks (66 statements, 5,206 numbers, 1,902 of them from 10-Ks); *set 4*, new for qcore9 (*set D* in the result files), 24 further filers, 16 10-Qs and 8 10-Ks (72 statements, 5,351 numbers, 1,928 from 10-Ks), built exactly as set 3. Truth is each filing's XBRL instance located on the page; identities come from the filers' calculation linkbases (set 3: 640 of 1,076; set 4: 998 of 1,135) plus semantic and layout rules, and all hold on the truth. Poor scans: 110 dpi, up to 1.5° tilt, blur, noise, specks, contrast 0.8, JPEG quality 25. Hand identities cover 82.8% of the numbers of sets 1–3 and 85.5% of set 4. **CORD receipts.** Test split, 100 receipts, 998 numbers, poor copies at 144 dpi; coverage 71.5%. **Notes tables, clean pages.** 233 dense notes and MD&A tables of 30 held-out filers (14,778 numbers) at 150 dpi; 76 tables of 10 other filers are development data. Identities cover 32.5% of the numbers of the 309 tables. **Real scans.** 77 held-out VRDU ad-buy forms with annotated amounts (1,009 numbers), faxes and scans; on 16 of them the annotated lines do not sum to the annotated gross. **Synthetic domains.** Held-out bundles 02–05 of quarterly corporate reports, invoices and statistical bulletins (192 documents, 7,892 numbers), with 24 planted source-document issues that are correct as printed. **Fresh synthetic sets (qcore7 and qcore8 held-out).** qcore7's: nine new bundles of reports, invoices and bulletins from the same generator (144 documents, 5,920 numbers). qcore8's: 18 bundles, 199 documents, 13,868 numbers; 12 of them (103 documents, 9,860 numbers) from four generators written for that test, whose layouts and arithmetic no version was tuned on (budget-vs-actual reports, hospital and school census, inventory logs and roll-forwards, utility bills tied to bank statements). No model read them: each has three extractions made from the true values, *clean*, *low* (about 1.5% injected errors) and *high* (5%): digit look-alikes, transpositions, dropped digits and decimal points, column and row shifts, sign flips, blanks. Text and no-text modes only. **Models.** Claude Opus and Claude Haiku read every set from page images; GPT (gpt-6-luna) read the SEC scans and the notes tables. A GPT blank on SEC always counts as wrong; elsewhere a blank is right only where the truth is 0. On sets 3 and 4 Haiku and Opus read the poor-scan pages as subagents, six statements per agent, with the same prompt (set 4: 2 Haiku blanks, 0 Opus, 10 GPT). The prompt asks for each number as printed and does not forbid using the printed totals to settle an unclear digit, and the models do: of the notes they left on set 4 (Haiku on 67 statements, Opus on 52), 40 and 38 mention footing, many recording a digit chosen so that a total foots. Such a digit ties every identity it feeds, so only the page can show it wrong (Section 5.9).

Table 1: Frozen versions, what each was tuned on, and what it was run on once. Earlier versions are in Appendix J.

Version	Frozen (Oct. 2026)	Tuned on	Tested once on
qcore4	15th 18:33	SEC dev filings: poor scans, OCR, injected errors; CORD train, val.	SEC 1, 2; CORD; notes; real scans
qcore5	16th 03:28	+ notes tables of the 10 dev filers; synthetic dev; PE validation	the same, all modes, three models
qcore6	16th 13:14	+ real-scan dev split; real re-reads of dev crops (Haiku, Opus readers)	SEC 1, 2, 3 (new); CORD; notes; real scans; synthetic 02–05
qcore7	16th 16:40	+ every qcore6 held-out set (now spent); zero-config only	fresh synthetic 06–08, injected errors
qcore8	16th 21:54 (3rd freeze)	+ qcore7's fresh set (now spent); zero-config only	4 new document families; old generators 09–10; injected errors
qcore9	18th 18:27 (1st freeze)	+ qcore8's fresh set (now spent); zero-config, and scaled page matches in every mode	SEC 4 (new): 24 filings, with qcore8, qcore6

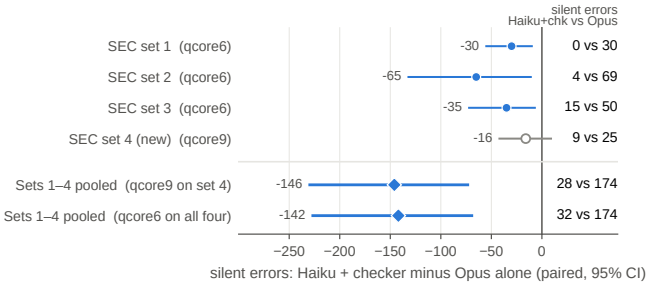


Figure 3: Haiku with the checker (text layer, hand identities) against Opus alone, per held-out SEC set and pooled: paired difference in silent errors with its 95% bootstrap interval; right, the two counts. Every interval lies below zero except set 4's, where Opus alone made only 25 errors (hollow: not significant).

4.3 Scoring

A number is *silent* if it is wrong at the end and named by no tier 2 item, *flagged* if a tier 2 item names it, and *broken* if the model had it right and the checker made it wrong. "To a person" is the flagged share of all numbers. Intervals are 95% percentile bootstraps over documents (10,000 resamples); paired differences use the same resamples, and a difference is significant when its interval excludes 0. A value is right when $|v-t| \le \max(0.5, 10^{-6}|t|)$, the rule of every earlier report. Set 4 is scored with this rule, for comparability with sets 1–3, and also with a *strict* rule, right within half a unit of the printed number's last digit, which also sees errors in per-share amounts: it breaks the same numbers and gives the same headline, 9 against 25 (Appendix E). The qcore7 and qcore8 synthetic runs are scored primarily with the strict rule.

5 Results

Figure 4 and Table C.1 (Appendix C) give every qcore6 held-out set, model and mode with hand identities (SEC sets 1 and 2 separately and all paired differences in Supplement S2); Figure 3 and Table E.1 (Appendix E) give set 4 with qcore9 and the four SEC sets pooled. The long per-run listings (every run, every paired difference, the development grids) are in a separate supplement, `supplement.pdf`, built from the same result files.

5.1 A cheap model with the checker against the strongest model alone

Set 3, frozen qcore6. On set 3 Opus alone is 99.04% accurate and leaves 50 silent errors (21–89); Haiku alone leaves 216. With qcore6 reading the born-digital text layer, Haiku ends at 98.21%: the checker fixed 123 of Haiku's errors from the statements' own lines, flagged 78 more (3.3% of numbers to a person) and left 15 silent (5–28), breaking nothing. The paired difference against Opus alone is −35 (−73 to −6). Opus with the checker leaves 5 (0–12) of its 50, a cut of −45 (−82 to −17). GPT goes from 1,443 silent errors to 65 (fixing 887), not distinguishable from Opus alone (+15, −32 to +58), but it broke two numbers (Section 5.2).

Set 4, frozen qcore9 (Figure 3, Table E.1). On the 24 newest filings Opus alone is 99.53% accurate and leaves 25 silent errors (7–49), half its count on set 3; Haiku alone leaves 168 and GPT 1,090. With qcore9 reading the text layer, Haiku leaves 9 (0–26) at 1.9% of numbers to a person, with 124 fixes and no correct number broken. Against Opus alone that is −16 (−43 to +10): fewer, but not significantly; a set on which Opus errs 25 times gives the comparison little power. Strict scoring gives the same 9 against 25. Opus with the checker leaves no silent error (−25, −49 to −7), GPT 15 (−1,075), not distinguishable from Opus alone (−10, −38 to +17), with two numbers broken (Section 5.2).

Pooled. Over the 227 statements of sets 1–4 (17,125 numbers), with qcore9 on set 4 and qcore6 on sets 1–3, Haiku plus the checker leaves 28 silent errors (10–50) against Opus alone's 174 (102–258), a paired difference of −146 (−231 to −72), at 2.5% of numbers to a person and 0 broken. With qcore6 on all four sets it leaves 32 (14–55), −142 (−228 to −68). Opus with the checker leaves 5; GPT 115 (126 with qcore6 throughout) from 4,076, not distinguishable from Opus alone. Per set: 0 against 30 on set 1, 4 against 69 on set 2, 15 against 50 on set 3, all significant, and 9 against 25 on set 4.

Hand mode is no longer version-independent. qcore6, qcore7 and qcore8 returned the same values with hand identities on every earlier set, and qcore8 and qcore6 still do on set 4, in all 12 hand runs. qcore9 returns their fixes and tier 2 items in the three no-text runs and in Opus's text run. Elsewhere its change 3 withdraws loose scaled page matches that had corroborated wrong values or held back right ones: Haiku with the text layer gets 4 more tier 2 items and its silent errors fall from 13 to 9 (−4, −8 to −1); each OCR run gets 1 to 5 more and loses at most one silent error; GPT with the text layer gets 3 more fixes, 14 tier 2 items more and 3 fewer, and falls from 26 to 15 (−11, −23 to 0), but breaks one more number (Appendix E).

Baselines (sets 1 and 2, from an earlier version's runs on the same extractions): flagging every input of a failing identity leaves 142 silent errors; re-asking Opus on every statement with a failing identity, plus those flags, 75, breaking 25 numbers Haiku had right; using the text layer itself as the extractor, Haiku filling the rest, plus flags, 36 (20–54) at 5.0% flagged, a baseline flattered because the truth was located on that layer. Haiku plus the checker leaves 4 there.

5.2 Every model gains; what broke

Over the 60 per-set (set, model, mode) settings of qcore6 with hand identities, the checker cuts silent errors with an interval below zero in 50, and in 41 of those it broke no correct number. On set 4 qcore9 does so in 11 of 12 hand settings, 8 with no break; the exception is Opus without page text (25 to 17). Counting discovered identities and zero-config, 26 of its 30 set-4 settings cut significantly; the four exceptions are Opus without page text. The ten exceptions are settings with almost nothing to cut: Haiku and Opus on the clean notes tables, the real scans with the scanner's OCR layer (16 and 15 silent errors fall to 1, interval touching zero), and the synthetic

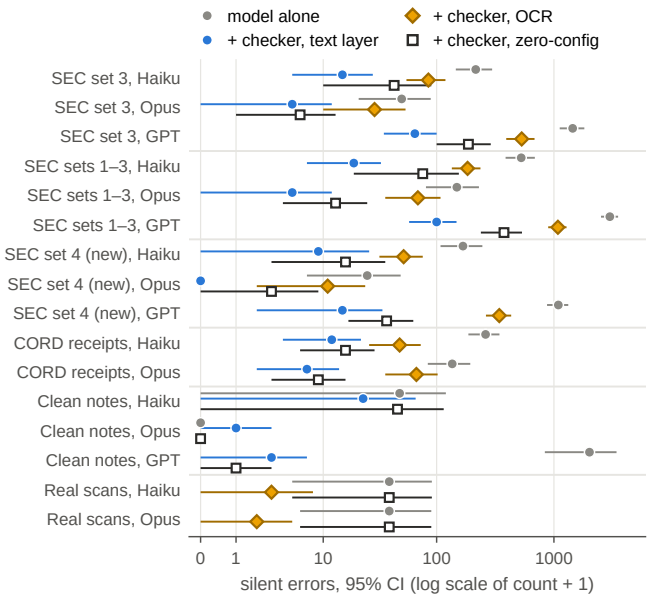


Figure 4: Silent errors per held-out set and model, alone and with the checker (frozen qcore6; set 4: frozen qcore9), with 95% bootstrap intervals; log axis of the count plus one, so 0 is the left edge. *Text layer* and *OCR* (docTR; real scans: the dataset's OCR) use hand identities; *zero-config* uses none (real scans: no page text). With the text layer the checker cuts silent errors by one to two orders of magnitude; OCR cuts far less. Counts: Tables C.1 and E.1.

bundles, where the models make 0 to 2 errors. On CORD the checker cuts Haiku from 262 silent errors to 12 and Opus from 136 to 7 with the text layer, at 29.2% and 17.0% of numbers to a person: receipts have few identities per number, so the checker turns errors into a review queue rather than repairing them.

Twelve distinct correct numbers were broken, all silently (Table C.4). Nine came from OCR mode, where a one-glyph look-alike made an identity tie (for example a Walmart current debt, 3,542 to 4,542 M, and an Expeditors deferred-tax line, 3.04 to 4.04 M). Two are a pair on the new filings: in GPT's text run on a Snap-on 10-K cash flow, the period-swap repair exchanged two numbers from different rows and years (22.8 and 23.3 M) that happened to tie the identities. One is a Salesforce debt line on the notes tables ($39,500 M to $39,288 M), broken for all three models by rule (b)'s "extracted number printed elsewhere on its row". No no-text run broke a number on any set, and no text run broke one on receipts, real scans or the synthetic bundles. With hand identities qcore7 and qcore8 return the same values, so these breaks are theirs too. On the synthetic held-out sets hand identities broke one more number, a budget figure (326 to 331) on qcore8's fresh set, identically in all three versions.

Set 4 adds three numbers broken with hand identities, all silent (Appendix E). *A text layer that misprints signs*: in WAT's PDF text layer the closing parenthesis is a separate word, '(2.31' then ')', so the page shows a basic loss per share of −2.31 as positive; with no identity through earnings per share, rule (b) took that read in GPT's text run, in every version. This is a tokenisation gap, not a qcore9 regression, and it causes every zero-config break on set 4 (Section 5.3). *A row shift, qcore9 only*: GPT shifted a 10-K's 2024 operating cash-flow column one row; qcore9's change 3 applied two correct page reads that qcore8 had withheld, the operating sum still failed from the rest of the shift, and identity completion then finished it on the one correct cell (WST other current assets, −5.8 to +5.7 M): net, 2 more fixes and 1 more break than qcore8. *OCR look-alikes*: in Haiku's two OCR runs, in every version, a Domino's and a Garmin balance-sheet line were changed to the one look-alike of the OCR read that tied an identity, which had failed because another of its cells carried a Haiku error, the class of set 3's OCR breaks.

5.3 Zero configuration

Table C.2 (Appendix C) runs the same data without the hand-written files. *Zero-config* (no identity file, no schema) still cuts silent errors against the model alone on every SEC and CORD row: pooled SEC, Haiku 527 to 76, Opus 149 to 13, GPT 2,986 to 376 in text mode; CORD, 262 to 16 and 136 to 9. It is significant in 24 of 34 per-set settings, and its only breaks are the eight synthetic bulletin cells of Table C.4. But it is weaker than the hand identities on every SEC row: pooled Haiku leaves 76 against 19 and fixes 142 errors against 348; GPT leaves 376 against 100 and fixes 155 against 2,020; and Haiku zero-config is no longer distinguishable from Opus alone (76 against 149, −73, −173 to +30). The inferred schema costs more than the discovered identities: with the hand schema and discovered identities only, pooled Haiku leaves 26 and fixes 345.

Two failure modes are clear. On the real scans without page text, discovery has nothing on the page to validate against: it finds identities covering about 41% of the numbers and they flag nothing, so the 39 silent errors stay (hand: 2 and 3). On the synthetic bulletins, discovered identities break 16 correct numbers per model (8 under zero-config), all in one age-band row, '30-34', whose label the page reader misreads and which the hand identities never let through; planted-issue routing falls from 22–24 to 8–12 of 24. Both failure modes were the targets of qcore7 (Section 5.4).

Adding the discovered identities to the hand-written ones (*hand + disc.*) raises coverage on pooled SEC from 82.8% to about 90% and on the notes tables from 32.5% to 61–67%. It was never significantly worse than hand identities alone in 46 comparisons and removed the Salesforce break on the notes tables.

Set 4: the final zero-config mode on real held-out filings (Figure 5). The qcore7–qcore9 zero-config changes had been tested held out only on synthetic data (Section 5.4). On set 4, with no identity or schema file, qcore9 leaves Haiku 16 silent errors (3–36) with the text layer, GPT 37 (17–63) and Opus 3 (0–9), against 9, 15 and 0 with hand identities and 168, 1,090 and 25 for the models alone. Haiku and GPT are significantly worse than with hand identities (+7, +1 to +15; +22, +10 to +35) but far below the model alone, and Haiku zero-config is not distinguishable from Opus alone (−9, −36 to +17). qcore8 on the same inputs left 19 and 51 (GPT: −14, −25 to −5) and qcore6 31 and 113. Discovered identities covered

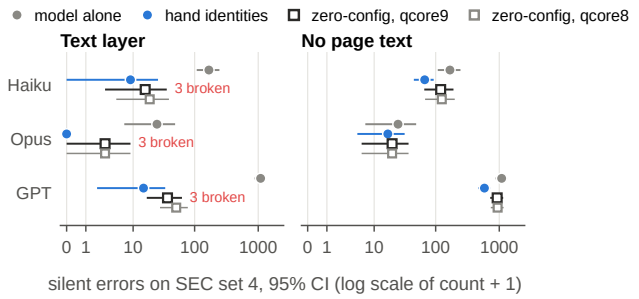


Figure 5: SEC set 4: the model alone, frozen qcore9 with hand identities, and zero-config (no identity or schema file) in qcore9 and qcore8; silent errors with 95% intervals. Zero-config stays far below the model alone but above hand identities, and every zero-config text run broke the same three numbers (WAT's text layer prints the signs wrong).

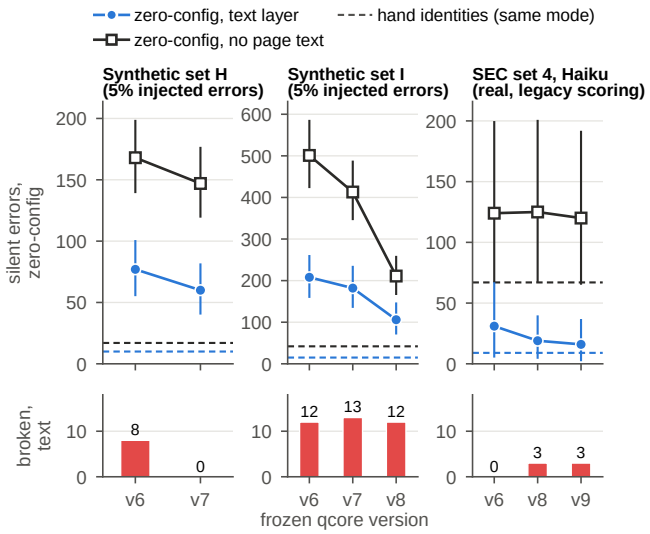


Figure 6: Zero-config across frozen versions, each run once on a set it was not tuned on. Top: silent errors (95% intervals) on the fresh synthetic sets H and I at the 5% injected-error rate (strict scoring) and for Haiku on SEC set 4; dashed, hand identities in the same mode. Bottom: correct numbers broken with the text layer. Each version lowered zero-config's silent errors, most on set I, but the gap to hand identities stays.

84.0% of the numbers for Haiku, 88.3% for Opus and 69.8% for GPT, against 85.5% for the hand files; GPT's errors leave fewer templates holding. Every zero-config text run, every model, qcore8 and qcore9 alike, broke the same three numbers silently: WAT's text layer shows three negative cash-flow totals ((85), (337), (174) M) as positive, the label read proposes sign flips, and discovered vertical sums, checked against the same misread page signs, let the flips tie. Hand identities refused them. Without page text zero-config breaks nothing but loses more: Haiku 120 against 67 with hand identities (+53, +9 to +112), GPT 927 against 585; qcore9's new discovery cut qcore8's 125 and 942 only by 5 and 15.

5.4 qcore7 to qcore9 on synthetic and development data

Synthetic held-out sets (Figure 6; Appendices D and G). On qcore7's fresh set qcore7 broke no correct number where qcore6 broke 8 with zero-config, all one age-band cell, and cut zero-config silent errors at a 5% error rate from 77 to 60 with the text layer (−17, −25 to −10). On qcore8's fresh set (Table G.1) zero-config qcore8 cut qcore7's silent errors by a third to a half in every setting with errors (5% error rate: 182 to 106 with the text layer, −76, −107 to −50; 413 to 211 without), but left 2 to 7 times those of hand identities (106 against 15; 211 against 42). About four fifths of that gap at the high rate lay on the four new document families, which hold 71% of the numbers, as did every zero-config break and most extra flags. It sent 7.0% of an error-free extraction's numbers to a person with the text layer (hand 2.4%), and broke 2, 2 and 12 correct numbers in text mode on the clean, low and high extractions: check amounts on one bank-statement layout, flipped by an inferred sign convention, and an inventory row read from its neighbour. These sets are synthetic: our generators, injected errors rather than model errors, no OCR mode, and hand identities complete by construction, which favours the hand side.

qcore9 on development data (not a test). qcore8's fresh set became development data, and qcore9's changes 1–4 are aimed at qcore8's breaks there (the bank-statement checks, the inventory row, a scaled page match behind the hand-mode budget break, a census day). The frozen development grid of the version run on set 4 (Appendix F; legacy scoring, strict on sets H and I) confirms this. Zero-config qcore9 breaks no development number, where qcore8 broke 16 on set I, and leaves fewer silent errors than qcore8 on every SEC set (set 3: 1,525 to 1,460) and on sets H and I (110 to 103; 404 to 375), with at most 0.3 points more numbers to a person; the notes tables gain one silent error (95 to 96). With hand identities qcore9 returned qcore8's fixes in 350 of 352 runs and its tier 2 items in 330; the 22 that differ all read page text, and there change 3 cut silent errors on SEC set 3 from 2,304 to 2,241 and removed the set-I budget break. The regression is the CAT break above, in discovered and hand + discovered text runs (3 numbers each), which led to the refreeze. The refrozen package (freeze 2, `c4401b94`) fixes the CAT break on development data, with no new break (`out/qcore9_work/dev_summary9.md`); it has not been tested on held-out data.

5.5 The re-read tier: a second read is not independent evidence

On development scans we asked real readers (Haiku and Opus) about crops of the unsure cells. The first gate let a re-read that agreed with an OCR digit misread change the value; it changed 11 correct SEC numbers, every one a glyph misread (16,384 to 15,384; 285,091 to 225,091) that the OCR and the reader shared. The readers repeat the extraction too: of 50 correctly placed cells where the extraction was wrong, a Haiku reader repeated the extraction's wrong number in 16

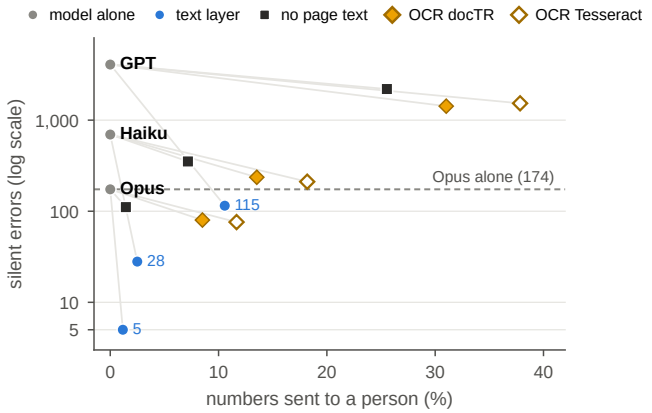


Figure 7: Silent errors against the share of numbers sent to a person, SEC sets 1–4 pooled (17,125 numbers; hand identities, qcore9 on set 4); thin lines join each model alone to its runs with the checker, and blue labels give the text-layer counts. Only the text layer takes the cheap model below Opus alone (dashed) at a small queue; OCR and no-text modes buy fewer silent errors with larger queues.

(32%) and an Opus reader in 8 (16%), and on 15 of Opus's 20 misreads Haiku misread the same cell. The hard glyphs on a poor scan are hard for every reader. So the frozen gate lets a re-read move a value only through an identity that then ties, and an agreeing re-read clears a flag only when the page read is another number altogether. On development scans the chosen Opus reader then cut Haiku's OCR-mode silent errors from 23 to 17 with no break.

Held out, on set 3 with docTR (Table C.3, Appendix C), an Opus reader answered 695 crops for Haiku's extraction and 523 for Opus's. On crops placed on the right cell it was right 89% and 85% of the time, but where Haiku's extraction was wrong it repeated the wrong number in 13 of 70 cells. The re-reads changed no value: the fixes, and the two OCR-mode breaks, are those of the run without them. They worked only through tier 2, adding 138 and 86 `reread_disagrees` items: Haiku's silent errors fell from 85 to 80 (-5, -10 to -1), Opus's from 29 to 27 (-2, -5 to 0), with 0.5 and 0.2 points more numbers to a person. That moved Haiku in OCR mode from more silent errors than Opus alone ($+35$, $+3$ to $+66$) to not distinguishable ($+30$, -3 to $+62$), not below it. The identities, not the extra read, carry the evidence. The re-read tier was not run on set 4.

5.6 Without a text layer the cheap model does not catch up

Figures 7 and 8 show the pooled SEC sets by page-text mode. When the only page text is OCR of the poor scan, Haiku plus the checker leaves 184 silent errors with docTR and 172 with Tesseract on pooled sets 1–3, against Opus alone's 149 (differences $+35$, -29 to $+91$, and $+23$, -45 to $+82$); on set 3 alone the docTR run is worse than Opus alone. With no page text it leaves 285, more than Opus alone ($+136$, $+73$ to $+202$). OCR mode fixes few errors (33 of Haiku's 527); its gain is routing. The checker still helps the strong model: Opus's silent errors fall from 149 to 69 with docTR (-80, -131 to -39) and 62 with Tesseract, at 9.2% and 12.4% of numbers to a person, and from 149 to 94 with the identities alone. Set 4 repeats this with qcore9: Haiku plus the checker leaves 67 silent errors without page text, 52 with docTR and 39 with Tesseract, against Opus alone's 25, significantly more ($+42$, $+16$ to $+68$; $+27$, $+2$ to $+53$) or not distinguishable ($+14$, -7 to $+34$); Opus falls from 25 to 11 and 14 with OCR (significant) and to 17 without page text (not significant). Pooled over sets 1–4, Haiku leaves 352 without page text (Opus alone 174; $+178$, $+109$ to $+251$) and 236 and 211 with OCR, not distinguishable from Opus alone. Feeding OCR to an earlier version as if it were a text layer broke 93 (Haiku) and 102 (Opus) correct numbers on set 1 (Appendix I); the two-reader rule removed that.

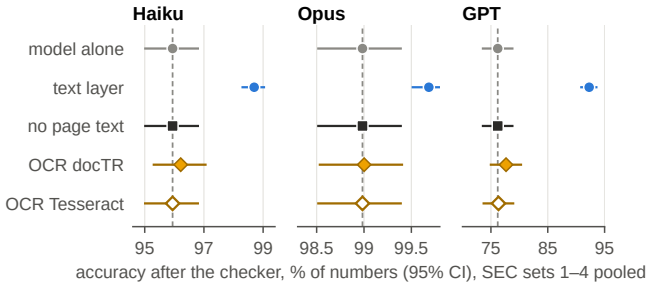


Figure 8: Accuracy after the checker by page-text mode, SEC sets 1–4 pooled, with 95% intervals; dashed, the model alone. Fixes come from the text layer; without page text the checker writes nothing, and OCR moves it by at most one and a half points: there the gain is routing, not repair.

5.7 Clean pages, real scans and synthetic documents: safety

Clean notes tables. Opus alone has no error on the 233 held-out tables; in no-text mode the checker adds neither flags nor damage, and in text mode it flags 2.1% and breaks the Salesforce cell. Haiku's 48 errors, structural slips such as shifted columns, fall to 23 or 24, but the cut's interval touches zero: half of them lie outside every identity. GPT's 2,005 errors are 2,001 blanks; the checker names every blank and leaves 3 silent at 18.0% of numbers to a person, a router for a model that abstains. **Real scans.** Both models leave 39 silent errors on the 77 forms, all in the 16 forms whose annotations fail their own sums. In every run the checker changed no number; with no page text or the dataset's OCR it routed 13 (Opus) or 14 (Haiku) of the 16 and left 2 or 3 silent at 13.8–17.4% of numbers to a person. On 33 amounts in six forms both models read the same value against the annotation, which points to annotation errors; we did not inspect the images to settle this. **Synthetic domains.** Opus makes no error on 7,892 numbers and Haiku 2; with hand identities the checker changes no correct number, changes none of the 24 planted source-document issues, and routes 22 (Haiku) and 24 (Opus) of them.

Table 2: Correct numbers broken by tier 1 in text mode on held-out data, by the version not tuned on that kind of document and by the version tuned on its development split (per extractor for CORD, SEC and notes).

Held-out data	Versions	Numbers	Untuned	Tuned
Synthetic domains, 24 runs	qcore, qcore2	15,784	164	32
CORD receipts	qcore2, qcore3	998	157–192	0
SEC set 1, clean	qcore2, qcore3	2,228	9	0
SEC notes tables, clean	qcore4, qcore5	14,778	249–292	1

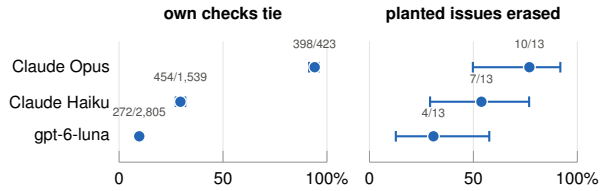


Figure 9: Bending to tie on the private-equity held-out set 2 (bad scans, five funds per model), with Wilson 95% intervals. Left: share of tier 0 errors whose checks all pass on the model's own numbers. Right: share of the 13 planted document issues per model that the model's entries erased.

5.8 What made the page read safe

Every time a frozen version met a new kind of document without a development step, its page reader broke correct numbers and its no-text mode broke none (Table 2). qcore4 applied any own-line read that broke no tied identity, and on the notes tables, where few numbers are tied and wide tables stack period blocks over segment columns, it broke 292 of Opus's held-out numbers. The gate of Section 3.2, tuned on development tables only, cut that to the one Salesforce cell. The same lesson returned with re-reads: agreement between two readers is not enough where their errors are correlated. A change needs an identity behind it, or a cell that cannot be confused with another.

5.9 Models bend numbers so totals tie

The checker grew out of one for limited-partner packets from private funds (Appendix A), where with a text layer the frozen PE checker fixed 4,685 of 4,767 model errors on held-out synthetic funds and changed no correct number. There we first saw the models bend numbers. Opus wrote: "Where a digit was unclear, I chose the value that makes the statement's own totals add up". On the held-out funds 398 of Opus's 423 errors tie every identity they feed (94.1%), against 29.5% of Haiku's, and Opus erased 10 of 13 planted source-document errors by entering consistent values (Figure 10 shows Haiku making the same move on a synthetic fund). Its notes on the SEC scans read the same way: "2026 accounts receivable: read (368) or (388). Chose (388) so operating components foot"; the page prints (368). A check that only sums the extraction cannot see such a number, which is why no-text and OCR modes leave tens of Opus's errors silent, and why rule (3) forbids the checker the same move. Figure 9 shows both measures for the three models.

6 Discussion

A recipe. For a document set whose numbers obey linear identities, run a cheap model, then the checker; send its packets to a stronger model or a person. When the PDF keeps a born-digital text layer, this leaves fewer silent errors than the strongest model alone, at a review queue of a few percent. When the page text is OCR, run the checker behind the strongest model: it cuts silent errors by routing, but a cheap model will not catch up. On clean pages with the strongest model there is little to fix, and no-text mode costs nothing. On sparse-identity documents such as receipts, expect a review queue of 9 to 31% of numbers.

Cost of a new domain. Writing the identities was the main cost of each earlier domain. Zero-config removes it at a price: on SEC statements it keeps most of the cut against the model alone but loses most of the fixes, and it needs page text to validate what it finds. qcore7 made zero-config more cautious and qcore8 and qcore9 stronger; on new real filings qcore9's zero-config left Haiku 16 silent errors where hand identities left 9. But on fresh synthetic layouts it still leaves several times the silent errors of hand identities and sends more numbers to a person, and each inference step (a tolerance, a label, a sign convention, a page sign) has opened a new path to a wrong change. In practice we would start zero-config, read the discovered identities (each run lists them with their support), add a hand file for the important ties, and run hand plus discovered.

What a second opinion is worth. The re-read result generalises beyond our checker: two vision models reading the same degraded pixels share their hard glyphs, so agreement between them is weak evidence exactly where it is needed. An identity is evidence of a different kind, because it does not look at the cell at all.

7 Limitations

Text-layer dependence. The cheap-model result needs a born-digital text layer behind the image, and that layer can be wrong: WAT's splits a closing parenthesis from its number, which caused every zero-config break on set 4. With OCR, Haiku plus the checker is not distinguishable from Opus alone on the pooled statements and is worse on sets 3 and 4 with docTR; with no page text it is worse. For CORD, text mode reads an OCR annotation that matches the truth by construction, and for SEC and the notes tables the truth was located on the same text layer that text mode reads; no-text and OCR rows are free of this.

Identity coverage. An error in a number no identity covers can be caught only by the page read. On the notes tables hand identities cover a third of the numbers and half of Haiku's errors lie outside them; discovered identities double the coverage there but did not reduce silent errors significantly.

Power on set 4. Opus alone made only 25 errors on set 4, so no comparison against it there can reach significance unless the checker's pipeline is nearly perfect; the cheap-model claim

rests on set 3 and the pooled sets.

Breaks. With qcore6 and hand identities twelve correct numbers were broken on sets 1–3 and the other real sets, all silently: nine by OCR look-alike repairs, which identities alone cannot tell from a coincidental tie, two by a period swap that tied by accident, and one by a page read. On set 4 qcore9 with hand identities broke three: a misprinted page sign accepted by rule (b), a GPT row shift that change 3 helped identity completion finish on a correct cell (new in qcore9), and OCR look-alikes in Haiku's OCR runs; with zero-config it broke three cash-flow totals per model, every version alike, from the same misprinted signs. Zero-config qcore6 broke eight synthetic cells on a digit-only label; qcore8 broke 2 to 12 per extraction on its fresh synthetic set, through an inferred sign convention and an ambiguous row label; hand identities broke one synthetic budget figure in qcore6–qcore8.

Extraction protocol. The extraction prompt does not forbid choosing an unclear digit so that the printed totals foot, and the models' notes show they did. That lowers the errors identities can see without the page, on every SEC set alike; text mode is not affected, since it compares each cell with the page.

The qcore7 and qcore8 tests are synthetic: our own generators, injected errors, no OCR mode, and hand identities complete by construction; qcore8 was frozen three times after development breaks before its run. The gap to hand identities is not closed (2 to 7 times on qcore8's fresh set, mostly on new layouts; 16 against 9 for Haiku on set 4), and zero-config qcore8 sends 7.0% of an error-free extraction's numbers to a person, about three times the hand identities' share.

Which qcore9. The set-4 results are those of qcore9's first freeze. The package now carries a second freeze, made after that run to fix a development break in discovery; it was not run on set 4. Its development grid found no new break; it describes its two freezes as both preceding any held-out run, which the time stamps contradict for the second.

Re-reads. One reader model, one OCR engine and one set were tested held out, with answers written once; crop placement follows the OCR, and 18–20% of crops sat on another cell. Set 4 has no re-read tier.

Scope. Three models from two providers; GPT was not run on CORD, the real scans or the synthetic bundles. One degradation recipe for the SEC and CORD scans. The real-scan truth fails its own sums on 16 of 77 forms. Tier 2 was tested only with re-asks of Opus. SEC set 1 was seen before qcore4 was designed. We downloaded SROIE receipts but did not evaluate on them.

8 Conclusion

A cheap model followed by a deterministic checker that reads the page and the documents' own identities left 15 silent errors on 22 new filings where the strongest model alone left 50, 9 against 25 on 24 further filings, run once with a version frozen before it, and 28 against 174 over 227 held-out statements; on the last set alone the gap is not significant, because the strong model erred little. Over eight held-out sets and three models the checker cut silent errors in most settings, and every number it broke is listed with its cause. It can find its identities on the page with no configuration, at a measurable cost: on the newest filings that mode left 16 silent errors for the cheap model where hand-written identities left 9, and broke three numbers per model where the filing's own text layer misprints a sign; on synthetic sets it left 2 to 7 times as many. A second model reading the same cell is correlated with the first; the identities are not, and so the rule that keeps the checker safe stays narrow: change a number only when an identity backs the page or the cell cannot be mistaken, write only what the page prints or what was extracted, and send the rest to someone who can look.

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Q1 2025 capital account, synthetic fund 37, bad_scan (85 dpi). Columns: ITD, YTD, QTD.

1 Tier 0 (Haiku). Unrealized gain ITD = **256,477**;
carried interest ITD = −10,939.
The model's note on the carry entry: "low confidence, blurred".

2 Q1 ITD roll-forward ties. $710{,}000 - \cdots + 256{,}477 - 10{,}939 = 913{,}757$ ✓
The −40,000 and +40,000 cancel, so a check inside the quarter passes.

3 Cross-quarter identity fails. $u^{\text{ITD}}_{\text{Q2}} = u^{\text{ITD}}_{\text{Q1}} + u^{\text{QTD}}_{\text{Q2}}$:
$256{,}477 + 218 = 256{,}695 \neq 257{,}995$, gap 1,300 ✗

4 Label read. Each field's own line prints the value: **296,477** (ITD),
carry −50,939; Q2 unrealized 618 (QTD) and 297,095 (ITD).

5 Tier 1 writes the printed values.
$256{,}477 \to 296{,}477$; $-10{,}939 \to -50{,}939$;
Q2: $218 \to 618$, $257{,}995 \to 297{,}095$
$296{,}477 + 618 = 297{,}095$ ✓ roll-forward still 913,757 ✓
every roll-forward and accumulation check on the capital accounts ties

Figure 10: A worked repair on a synthetic fund (held-out fund 37, bad scan, Claude Haiku; frozen v2 checker). The model entered an inception-to-date unrealized gain 40,000 below the printed value and a carried-interest charge 40,000 smaller, so the quarter's roll-forward still ties. The cross-quarter accumulation identity fails, and the fields' printed lines supply a fix that ties it; after tier 1 every roll-forward and accumulation check on the capital accounts ties.

A Private-equity case study: details

This appendix keeps the private-equity (PE) work the general checker grew out of, summarised in Section 5.9. "v1" and "v2" are PE v1 and PE v2 of Table J.1. Planted-error benches, a second real fund and planted Fund R results from the PE-only version are not repeated here.

Each later version was checked against earlier data: qcore reproduces PE v2 on the fund held-out set with no wrong change, and qcore2, qcore3 and qcore4 give identical per-error outcomes on all fund errors. On the PE validation runs qcore5 fixes 4,687 of 4,767 errors where qcore4 fixed 4,718 (37 fixes became tier 2 items, 6 the reverse), with no wrong change and no broken number in either.

Data. Fund R is one real venture fund's packet (name withheld): 13 PDFs, 59 pages and 252 numbers, with 6 real document inconsistencies. We degraded it to image-only scans at four levels (clean 100 dpi; scan 150 dpi; bad_scan 85 dpi with tilt, noise, specks and JPEG quality 18; headless, which is bad_scan with column headers erased), and Claude Haiku, Sonnet and Opus extracted it, 10 runs in all. The six bad_scan and headless runs contain 193 errors. Synthetic funds come from a quarter-by-quarter fund simulator; fifteen hard funds were degraded at bad_scan and extracted by Haiku, Opus and gpt-6-luna, in a development group and two held-out sets.

With a text layer. On Fund R the frozen PE v1 fixed 189 of 193 hard-run errors (97.9%), sent 2 to tier 2, missed 2 and made no wrong change. On held-out set 2 (funds 31–39), the frozen PE v2 fixed 4,685 of 4,767 errors (98.3%, Wilson 95% interval 97.9–98.6%), sent 81 to tier 2 and missed 1. It wrote no wrong value and changed no correct number: 0 of 4,767 errors and 0 of 5,349 correct values, a one-sided 95% upper bound of 0.06% each. Opus alone leaves 423 of 3,372 numbers wrong there; Haiku plus tier 1 leaves 15, of which 14 are in tier 2 packets (Figure S1.1). Without page text, v2 flags 75.3% of these errors but only 24 of Opus's 423, because Opus's errors tie its own identities (Figure 9). An ablation on the six Fund R hard runs shows where the fixes come from (Figure S1.2): without the label read, tier 1 fixes 14 errors instead of 189, and arithmetic alone fixes 5, routes 124 and misses 64.

With OCR instead of a text layer. When we replaced the Fund R text layer with Tesseract OCR of the degraded pages, with two safeguards (pages with median word confidence below 60 count as blank, and a number is kept only if a digits-only re-read agrees), tier 1 fixed 8 of 197 errors over nine runs and flagged 156, with no wrong change (Figure S1.3). At 85 dpi digits are about 6 px tall.

Under attack. A suite of 360 constructed cases (coordinated bending, decoy numbers, coincidental ties, planted source errors, label collisions, sign and scale traps, OCR-style digit confusions, blanks) made the frozen PE v1 write wrong values in 43 cases, all from page reads. PE v2, tuned on that suite, wrote no wrong value in either mode on 430 fresh cases generated after its freeze (upper bound 0.69% of cases), where v1 wrote wrong values in 33 (Supplement S1.5). The lesson carried into the general checker: a page read must answer to the identities (the guard of Section 3.2) just as a repair does (2).

Bending. On fund held-out set 2, 398 of Opus's 423 errors tie every identity they feed (94.1%), against 454 of Haiku's 1,539 (29.5%) and 272 of luna's 2,805. Each planted document issue fails its identities when the page is copied exactly, so if those identities tie in the extraction, the model has erased a real error in the source. Opus erased 10 of 13 planted issues (Wilson 49.7–91.8%), Haiku 7 and luna 4. Per-run tables of the PE work (Fund R and synthetic-fund runs, label-read failure causes, planted errors, adversarial results per category, erased document issues), the five v1 defects with their v2 changes and four further figures are in Supplement S1.

Table A.1: PE evaluation stages, in order

Funds 11–19 and the 360-case suite were test data for v1 and tuning data for v2. Funds 31–39 were built (18:13 UTC) and scored with v1 (18:35) before the last v2 source edit (18:52) and the v2 freeze (19:19); the change log and the tuning outputs show that v2 was tuned only on the 360 cases and the v1 regression runs, and the fresh attack cases were generated after the freeze. This is evidence rather than enforcement.

Stage	Data	Code; role
1 Development	Fund R runs; funds 01–09	pre-freeze, then v1 frozen
2 Held-out 1	funds 11–19, three models	v1; first test of v1
3 Adversarial	360 cases on funds 11–19	v1; found five defects
4 Fixes	the 360 cases plus v1 regression runs	tuning, then v2 frozen
5 Held-out 2	funds 31–39, three models	v2 (v1 also run); test of v2
6 Fresh adversarial	430 cases on funds 31–39	v2 (v1 also run); test of v2

B Synthetic domains

Table B.1: Zero-shot qcore over all 18 synthetic-domain bundles

Opus and Haiku read every bundle; gpt-6-luna read 16 of the 18. No-text mode broke nothing.

Model	Runs	Numbers	Tier 0 errors	Broken, text	Broken, no text	Tier 2 items, text
Claude Opus	18	11,983	1	123	0	318
Claude Haiku	18	11,983	4	123	0	303
gpt-6-luna	16	10,515	13	104	0	288

Table B.2: Held-out bundles 02–05 (Opus and Haiku, 24 runs per mode): zero-shot qcore against frozen qcore2

Checker	Mode	Numbers	Tier 0 errors	Broken	Tier 2 items	Planted routed	Planted changed
qcore2	text	15,784	2	32	134	46 / 48	0
qcore	text	15,784	2	164	414	47 / 48	0
qcore2	no text	15,784	2	0	83	46 / 48	0
qcore	no text	15,784	2	0	85	46 / 48	0

By domain, qcore2 in text mode broke 0 numbers on the quarterly reports and the invoices and 32 on the bulletins, where qcore broke 0, 24 and 140.

C Held-out results of frozen qcore6: every set

Frozen qcore6 (28 files, combined sha256 `86debea05f95a6d2d93a7add48d32cd54e6109135052df4c1dd178c48a95ab34`, frozen 2026-10-16T13:14:47Z; the hash matched before the runs, at their end and at report time) was run once per held-out set, model, mode and identity setting, with frozen qcore5 re-run on the same inputs: 442 runs, none crashed, none repeated, and in every run the returned values equal the extraction plus the tier 1 fixes. The qcore5 re-run reproduces its earlier held-out results row for row, and qcore6 with hand identities returns exactly qcore5's fixes and tier 2 items in all 104 (set, model, mode) runs, so every qcore6-against-qcore5 paired difference is 0; none of qcore6's default-path changes fired, because every text-mode PDF here has a born-digital text layer. On the notes tables and the real scans the checker ran over the whole bundle and rows are scored on the held-out documents. SEC set 3's extractions were checked against their locked hash manifests (132 files) before scoring, and its page images are pixel-identical to those the models read. Sources: `out/qcore6_heldout/results.md`, `results.json`, `SUMMARY.md`.

Identity settings. *Hand*: the bundle's `identities.json` and `schema.json`. *Discovered*: identities discovered from the documents, hand schema. *Hand + discovered*: both. *Zero-config*: discovered identities and an inferred schema, neither file read. OCR modes ran with hand identities only.

Re-read runs. Requests were generated once for set 3, OCR mode (docTR), hand identities, for the Haiku and Opus extractions (695 and 523 requests from 914 and 634 selected cells; the rest could not be placed on a row that prints the record's label, or shared a crop). One Opus reader answered every request once (29 and 53 null answers); the answer files are hashed in `SUMMARY.md`. Outcomes: Haiku extraction, agrees 515, disagrees 138, another cell 56, null 32, cleared 2; Opus extraction, 377, 86, 41, 57, 2. Over all reads (crops sharing a cell counted per record), the reader gave the truth 551 of 741 times (Haiku extraction) and 386 of 561 (Opus extraction); of its errors, 12 of 158 and 3 of 118 equal the extracted value and 68 and 64 equal the OCR's page read, mostly on crops placed on another cell, where reader and OCR read the same wrong cell.

Development data for the re-read gate (`out/qcore6_work/dev_summary6.md`). SEC development poor scans, Haiku extraction, OCR docTR (silent / flagged / fixed / broken): no re-read 23 / 438 / 14 / 0; first gate with real Haiku answers 30 / 370 / 15 / 4; final gate, Haiku

Table C.1: Every held-out set, frozen qcore6 with the bundle's own identities, one run per set, model and mode. Each cell: silent errors (wrong at the end and named by no tier 2 item) with a 95% bootstrap interval over documents, then the share of all numbers sent to a person (%). $^{\dagger n}$: the run broke *n* numbers the model had right. *Text*: the checker reads the PDF's born-digital text layer while the model reads the page image (CORD: the dataset's OCR annotation). *OCR*: docTR and Tesseract OCR of the same poor scan as a second reader; for the real scans, the dataset's OCR and the scanner's embedded OCR layer, the latter on the 57 forms that have one (there Haiku alone leaves 16 silent errors and Opus alone 15). –: not run. SEC sets 1 and 2 separately, and every paired difference, are in Supplement S2.

		Model alone		Model + qcore6: silent [95% CI] / % to a person			
Set	Model	Acc. %	Silent [95% CI]	Text layer	No page text	OCR 1	OCR 2
SEC set 3 (new): 22 filings, 15 10-Q and 7 10-K, poor scans of born-digital PDFs (66 statements, 5,206 numbers)							
	Haiku	95.85	216 [146, 298]	15 [5, 28] / 3.3	123 [82, 170] / 7.9	85 [55, 119]$^{\dagger 2}$ / 16.1	80 [53, 112]$^{\dagger 1}$ / 20.3
	Opus	99.04	50 [21, 89]	5 [0, 12] / 1.4	39 [13, 75] / 0.7	29 [10, 54] / 9.9	25 [10, 43] / 12.2
	GPT	72.28	1,443 [1,120, 1,813]	65 [35, 100]$^{\dagger 2}$ / 13.8	754 [582, 938] / 29.6	532 [393, 684] / 36.6	552 [423, 698]$^{\dagger 1}$ / 42.3
SEC sets 1–3 pooled, poor scans (155 statements, 11,774 numbers)							
	Haiku	95.52	527 [388, 688]	19 [7, 33] / 2.8	285 [213, 365] / 6.9	184 [135, 237]$^{\dagger 2}$ / 14.2	172 [127, 222]$^{\dagger 2}$ / 18.5
	Opus	98.73	149 [81, 230]	5 [0, 12] / 1.2	94 [50, 146] / 1.8	69 [36, 108]$^{\dagger 1}$ / 9.2	62 [31, 99] / 12.4
	GPT	74.64	2,986 [2,512, 3,509]	100 [58, 148]$^{\dagger 2}$ / 11.0	1,611 [1,362, 1,871] / 26.2	1,081 [894, 1,280]$^{\dagger 2}$ / 32.1	1,122 [940, 1,313]$^{\dagger 2}$ / 39.0
CORD receipts, poor scans of photos (100 receipts, 998 numbers)							
	Haiku	73.75	262 [187, 344]	12 [4, 22] / 29.2	52 [29, 78] / 25.1	48 [26, 73] / 29.0	46 [24, 71] / 31.1
	Opus	86.37	136 [84, 194]	7 [2, 14] / 17.0	67 [36, 102] / 8.7	67 [36, 102] / 12.7	66 [36, 100] / 14.8
SEC notes tables, clean pages (233 tables, 14,778 numbers)							
	Haiku	99.68	48 [0, 120]	23 [0, 66]$^{\dagger 1}$ / 2.3	24 [0, 68] / 0.2	–	–
	Opus	100.00	0 [0, 0]	1 [0, 3]$^{\dagger 1}$ / 2.1	0 [0, 0] / 0.0	–	–
	GPT	86.43	2,005 [837, 3,403]	3 [0, 7]$^{\dagger 1}$ / 19.7	3 [0, 7] / 18.0	–	–
Real scanned and faxed ad-buy forms (VRDU) (77 forms, 1,009 numbers)							
	Haiku	96.13	39 [5, 91]	not used	3 [0, 8] / 13.8	3 [0, 8] / 15.1	1 [0, 3] / 18.0
	Opus	96.13	39 [6, 90]	not used	2 [0, 5] / 16.6	2 [0, 5] / 17.4	1 [0, 3] / 18.0
Synthetic reports, invoices and statistical bulletins (192 documents, 7,892 numbers)							
	Haiku	99.97	2 [0, 5]	2 [0, 5] / 2.5	2 [0, 5] / 1.5	–	–
	Opus	100.00	0 [0, 0]	0 [0, 0] / 2.8	0 [0, 0] / 1.6	–	–

reader 19 / 449 / 14 / 0, Opus reader 17 / 449 / 14 / 0, both agreeing 19 / 445 / 14 / 0. Opus extraction: 2 / 259 / 0 / 0 without re-reads; first gate 5 / 247 / 0 / 7; final gate with the Opus reader 2 / 272 / 0 / 0. A simulated reader with independent 5% one-digit errors had suggested 12–14 silent errors and 22–23 fixes under the first gate: real readers are worse than independent noise because their errors follow the hard glyphs.

Natural errors on the clean notes tables (`notes_eval/scores/results.md`). Held-out: Opus 0 of 14,778; Haiku 48 (0.32%) in 3 of 233 tables, 24 of them in numbers covered by an identity; GPT 2,005, of which 2,001 are blanks, with 99.97% of its 12,154 non-blank answers right. Development: Opus 0 of 6,589; Haiku 16 sign errors in one table. Haiku's held-out errors are runs of values from the wrong place: a table whose columns it shifted (24 numbers), a table where every row got a different quantity (20), and four cells read from other rows.

Set 2 silent errors of Haiku plus the checker in text mode, the same four under qcore3 to qcore6: an ABBV share repurchase read as −1,496 for −1,498 M; a CVX share count 1,719,134 for 1,719,184; two Visa liability cells read from another row.

Real scans: amounts where both models agree against the annotation (`realscan_eval/scores/results.md`). 33 held-out amounts in six forms: rs_0030 (21), rs_0008 (5), rs_0009, rs_0026 and rs_0084 (2 each), rs_0078 (1). On the 61 held-out forms whose line items sum to the gross, both models agree with every annotation.

Table C.2: Where the identities come from. Silent errors, with identity coverage (share of numbers in some identity) in parentheses, for the bundle's hand-written identities, the discovered identities alone (hand schema), both, and zero-config (discovered identities and an inferred schema: no identity or schema file). $^{\dagger n}$: n correct numbers broken. Every row and setting is in Appendix H.

Set	Model	Mode	Alone	Hand	Hand+disc.	Zero-config
SEC 1–3	Haiku	text	527	19 (83)	19 (91)	76 (77)
SEC 1–3	Opus	text	149	5 (83)	5 (91)	13 (84)
SEC 1–3	GPT	text	2,986	$100^{\dagger 2}$ (83)	$95^{\dagger 2}$ (89)	376 (50)
SEC 1–3	Haiku	no text	527	285 (83)	283 (89)	421 (81)
SEC 1–3	Opus	no text	149	94 (83)	90 (90)	109 (87)
SEC 1–3	GPT	no text	2,986	1,611 (83)	1,587 (88)	2,482 (60)
CORD	Haiku	text	262	12 (72)	12 (78)	16 (50)
CORD	Opus	text	136	7 (72)	7 (79)	9 (59)
Notes	Haiku	text	48	$23^{\dagger 1}$ (32)	22 (67)	46 (56)
Notes	GPT	text	2,005	$3^{\dagger 1}$ (32)	0 (62)	1 (47)
Real scans	Haiku	no text	39	3 (99)	3 (99)	39 (41)
Real scans	Opus	no text	39	2 (99)	2 (99)	39 (42)
Synthetic	Haiku	text	2	2 (98)	2 (98)	$10^{\dagger 8}$ (81)
Synthetic	Opus	text	0	0 (98)	0 (98)	$8^{\dagger 8}$ (81)

Table C.3: The re-read tier on SEC set 3 (OCR mode, docTR, hand identities), held out. One reader (Opus, chosen on development data) answers a crop per uncertain cell. *On the cell*: the outlined cell is the record's own cell. Paired differences over documents, 95% intervals.

	Haiku extraction	Opus extraction
Requests (per 100 numbers)	695 (13.3)	523 (10.0)
Crops on the cell / reads	607 / 741 (82%)	448 / 561 (80%)
Reader right, on the cell	543 / 607 (89%)	379 / 448 (85%)
Extraction wrong, on the cell	70	12
reader gives the truth	45	5
reader repeats the extraction	13	3
OCR wrong / reader repeats the OCR	221 / 14	214 / 8
Values changed by re-reads	0	0
Silent, without → with	85 → 80	29 → 27
paired difference	−5 [−10, −1]	−2 [−5, +0]
To a person, without → with	16.1 → 16.6%	9.9 → 10.1%
Broken, with re-reads	2	0

Table C.4: Every correct number broken by frozen qcore6 on held-out data

Per-set runs only (pooled rows repeat them). Every break is silent (no tier 2 item names the cell). Reason: the tier 1 operation that made the change. The same cells broken under frozen qcore5 on the same inputs are not repeated.

Set	Pipeline	Cell (document | concept | period, column)	Truth	Changed to	Reason
SEC 1	GPT + qcore6, OCR docTR	VZ_2026-06-30_cash_flow | EmployeeBenefitsAndShareBasedCompensationNoncash | 2026-06-30 Six Months Ended June 30, 2026	35 M	30 M	misread
SEC 1	GPT + qcore6, OCR docTR	WMT_2026-07-31_balance_sheet | LongTermDebtCurrent | 2026-01-31 January 31, 2026	3,542 M	4,542 M	misread
SEC 2	Haiku + qcore6, OCR Tesseract	MCD_2026-06-30_cash_flow | ProceedsFromPaymentsForOtherFinancingActivities | 2025-06-30 Quarters Ended June 30, 2025	-126 M	-110 M	misread
SEC 2	Haiku + qcore6 discovered ids, text	MA_2026-06-30_balance_sheet | PropertyPlantAndEquipmentNet | 2026-06-30 June 30, 2026 (in millions, except per share data)	2,541 M	2,756 M	misread
SEC 2	Opus + qcore6, OCR docTR	HON_2026-06-30_cash_flow | DeferredIncomeTaxesAndTaxCredits | 2025-06-30 Six Months Ended June 30, 2025 (Dollars in millions)	-31 M	-50 M	misread
SEC 2	GPT + qcore6, OCR Tesseract	DIS_2026-06-27_cash_flow | IncreaseDecreaseInAccruedIncomeTaxesPayable | 2026-06-27 Nine Months Ended June 27, 2026	-1,611 M	-618 M	misread
SEC 3	Haiku + qcore6, OCR docTR	AES_2025-12-31_income_statement | GeneralAndAdministrativeExpense | 2024-12-31 2024 (in millions, except per share amounts)	-288 M	-256 M	misread
SEC 3	Haiku + qcore6, OCR docTR	ETSY_2026-06-30_cash_flow | IncreaseDecreaseInOperatingAssetsNoncurrent | 2026-06-30 Six Months Ended June 30, 2026	-3.239 M	-3.25 M	misread
SEC 3	Haiku + qcore6, OCR Tesseract	EXPD_2026-06-30_balance_sheet | DeferredIncomeTaxLiabilitiesNet | 2025-12-31 December 31, 2025	3.04 M	4.04 M	misread
SEC 3	Haiku + qcore6, OCR docTR + Opus re-read	AES_2025-12-31_income_statement | GeneralAndAdministrativeExpense | 2024-12-31 2024 (in millions, except per share amounts)	-288 M	-256 M	misread
SEC 3	Haiku + qcore6, OCR docTR + Opus re-read	ETSY_2026-06-30_cash_flow | IncreaseDecreaseInOperatingAssetsNoncurrent | 2026-06-30 Six Months Ended June 30, 2026	-3.239 M	-3.25 M	misread

Set	Pipeline	Cell (document | concept | period, column)	Truth	Changed to	Reason
SEC 3	GPT + qcore6, text	SNA_2026-01-03_cash_flow | AccountsReceivableAndContractReceivableCreditLossExpenseReversal | 2024-12-28 2024	22.8 M	23.3 M	wrong_period
SEC 3	GPT + qcore6, text	SNA_2026-01-03_cash_flow | IncreaseDecreaseInInventories | 2023-12-30 2023	23.3 M	22.8 M	wrong_period
SEC 3	GPT + qcore6, OCR Tesseract	POOL_2026-06-30_balance_sheet | AssetsCurrent | 2025-12-31 December 31, 2025 (Audited)	1,969.864 M	1,968.964 M	misread
SEC 3	GPT + qcore6 hand + discovered ids, text	SNA_2026-01-03_cash_flow | AccountsReceivableAndContractReceivableCreditLossExpenseReversal | 2024-12-28 2024	22.8 M	23.3 M	wrong_period
SEC 3	GPT + qcore6 hand + discovered ids, text	SNA_2026-01-03_cash_flow | IncreaseDecreaseInInventories | 2023-12-30 2023	23.3 M	22.8 M	wrong_period
Notes	Haiku + qcore6, text	CRM_2026-07-31_t036 | DebtInstrumentCarryingAmount | 2026-07-31 Outstanding Principal as of July 31, 2026	39,500 M	39,288 M	misread
Notes	Opus + qcore6, text	CRM_2026-07-31_t036 | DebtInstrumentCarryingAmount | 2026-07-31 Outstanding Principal as of July 31, 2026	39,500 M	39,288 M	misread
Notes	GPT + qcore6, text	CRM_2026-07-31_t036 | DebtInstrumentCarryingAmount | 2026-07-31 Outstanding Principal as of July 31, 2026	39,500 M	39,288 M	misread
Synthetic	Haiku + qcore6 discovered ids, text	public_stats_02 | narrative_count | entity 30-34, 8 quarterly bulletins	6,527–7,531	4,586–5,270	misread
Synthetic	Haiku + qcore6 discovered ids, text	public_stats_05 | narrative_count | entity 30-34, 8 quarterly bulletins	4,697–5,636	4,762–5,870	misread
Synthetic	Haiku + qcore6 zero-config, text	public_stats_05 | narrative_count | entity 30-34, 8 quarterly bulletins	4,697–5,636	20	misread
Synthetic	Opus + qcore6 discovered ids, text	public_stats_02 | narrative_count | entity 30-34, 8 quarterly bulletins	6,527–7,531	4,586–5,270	misread
Synthetic	Opus + qcore6 discovered ids, text	public_stats_05 | narrative_count | entity 30-34, 8 quarterly bulletins	4,697–5,636	4,762–5,870	misread
Synthetic	Opus + qcore6 zero-config, text	public_stats_05 | narrative_count | entity 30-34, 8 quarterly bulletins	4,697–5,636	20	misread

Synthetic rows (ranges over the bulletins): the record's entity is the age band '30-34', a digit-only entity; held-out qcore2 broke the same cells by the label read (Appendix B). The hand identities never let the read through; discovered identities tie it.
SEC sets 1, 2 and 3, CORD, the notes tables, the real scans and the synthetic bundles separately (every run, every identity setting), and every paired difference against the model alone, against Opus alone and between identity settings, are in Supplement S2.

D Frozen qcore7 on the fresh synthetic held-out set

Frozen qcore7 (28 files, combined sha256 `51d9a35236ca553c1bd1536cee0b06c344ce06c0f0020b467613c80eb0396e55`, frozen 2026-10-16T16:40:49Z, 65 tests) and frozen qcore6 were run once, 2026-10-16 16:41–16:46Z, on nine bundles built with the synthetic-domain generator, imported unchanged, with new seeds (base + 6, 7, 8): reports, invoices and bulletins 06–08, 144 documents, 5,920 numbers. The data were locked by a hash manifest (combined sha256 `6d6be8d8769f6181...`) at 15:27:55Z, before any qcore7 change; the two package hashes and the data hash were checked at the start and at the end of the run. 432 runs (2 versions × 9 bundles × 3 extractions × 2 modes × 4 identity settings), none crashed. The extractions are the true values (*clean*), and the true values with 89 (*low*) and 297 (*high*) injected errors; under strict scoring 89 and 296 numbers are wrong, under the legacy rule 81 and 278. Each extraction quotes the printed line of the PDF's text layer, as a model would. Sources: `out/qcore7_heldout/SUMMARY.md`, `results.md`, `results.json`; generator of these tables: `paper/gen_v7/gen_q7_tex.py`. Tables D.1 and S3.1 were in the main text of the previous version.

Table D.2: Every correct number broken on the fresh held-out set (strict)
qcore7 broke none in any of its 24 settings. Every qcore6 break is silent and is the age band '30-34' of the births bulletins (field `narrative_count`, one cell per quarterly bulletin), read by the label read as a value.

Extraction	Pipeline	Bundles (cells)	Broken	Changed to
clean	qcore6 discovered, text	public_stats_06 (8), public_stats_07 (8)	16	5,585–7,814
clean	qcore6 zero-config, text	public_stats_07 (8)	8	20
low, 1.5%	qcore6 discovered, text	public_stats_06 (8), public_stats_07 (8)	16	5,585–7,814
low, 1.5%	qcore6 zero-config, text	public_stats_07 (8)	8	20
high, 5%	qcore6 discovered, text	public_stats_06 (7), public_stats_07 (8)	15	5,725–7,814
high, 5%	qcore6 zero-config, text	public_stats_07 (8)	8	20

Every run, the paired differences, legacy scoring, injected errors by kind and by domain, and qcore7's development grid are in Supplement S3.

E Frozen qcore9 on SEC set 4 (set D)

Frozen qcore9 (30 files, combined sha256 `3ca510d3ea178550ee3a43676d9f8e74a4e4ec0c4e4e8b7d4149d72732d387e0`, frozen 2026-10-18T18:27:04Z, 79 tests; the package's first freeze, see below), frozen qcore8 and frozen qcore6 were run once, 2026-10-18 18:40–19:19Z, on SEC set 4 (`sec_eval4`): 24 filings (16 10-Qs, 8 10-Ks; ALB, CF, CLX, CTAS, DPZ, EXR, FAST, GRMN, HSY, JBHT, KMB, LKQ, LULU, MLM, NDSN, PAYX, ROL, STE, TER, TSCO, ULTA, WAT, WST, ZBRA), 72 statements, 5,351 located facts (USD 5,023, per share 207, shares 121), 1,135 identities (998 from calculation linkbases, for six filers read from the linkbase embedded in the schema file;

Table D.1: Frozen qcore7 against frozen qcore6 on the fresh synthetic held-out set (9 bundles of quarterly reports, invoices and statistical bulletins with new seeds, 144 documents, 5,920 numbers), one run. The extractions are the true values with injected errors (*clean*: none; *low*: 89 numbers wrong; *high*: 296 wrong); no model read these pages. Strict scoring (half a printed unit). Each cell: silent errors [95% bootstrap interval over documents] / % of numbers to a person (discovered identities with the hand schema: silent only); $^{\dagger n}$: n correct numbers broken. *Hand, both*: hand identities, where the two versions return the same fixes and tier 2 items in all 54 runs. Paired differences over the same resamples. Every row, the legacy scoring and every broken number: Appendix D.

				Discovered ids (silent)		Zero-config		Zero-config, paired silent diff.	
Extraction	Mode	Alone	Hand, both	qcore6	qcore7	qcore6	qcore7	qcore7 − qcore6	qcore7 − hand
clean	text	0	0 [0, 0] / 2.9	$16^{\dagger 16}$	0	8 [3, 14]$^{\dagger 8}$ / 2.0	0 [0, 0] / 2.4	−8 [−14, −3]	0 [0, 0]
	no text		0 [0, 0] / 1.4	0	0	0 [0, 0] / 0.6	0 [0, 0] / 0.6	0 [0, 0]	0 [0, 0]
low, 1.5%	text	89	5 [1, 10] / 5.5	$32^{\dagger 16}$	14	26 [17, 35]$^{\dagger 8}$ / 4.2	15 [8, 23] / 4.9	−11 [−18, −5]	+10 [+4, +16]
	no text		6 [2, 11] / 7.2	32	27	32 [21, 44] / 4.0	27 [17, 38] / 4.9	−5 [−11, 0]	+21 [+13, +30]
high, 5%	text	296	10 [4, 18] / 13.4	$77^{\dagger 15}$	54	77 [56, 100]$^{\dagger 8}$ / 7.7	60 [41, 81] / 8.5	−17 [−25, −10]	+50 [+32, +69]
	no text		17 [8, 27] / 20.5	168	147	168 [140, 198] / 7.7	147 [120, 176] / 8.8	−21 [−32, −11]	+130 [+104, +157]

Table E.1: SEC set 4 (set D in the result files): 24 new filings, 16 10-Qs and 8 10-Ks, 72 statements, 5,351 numbers, poor scans of born-digital PDFs; frozen qcore9, one run. Cells: silent errors (with the text layer, and for the model alone, with a 95% bootstrap interval over statements) / % of numbers to a person; $^{\dagger n}$: n numbers the model had right were broken. Hand: the bundle's identities and schema; ZC: zero-config, no identity or schema file (qcore8 on the same inputs in parentheses). Lower block: sets 1–4 pooled with hand identities; sets 1–3 are qcore6's frozen held-out runs re-scored (they reproduce its results in all 137 rows), set 4 is qcore6 (*all qcore6*) or qcore9; Opus rows are the same both ways. Legacy scoring. Paired differences: Table E.2; every run, with the qcore8 and qcore6 re-runs: Supplement S5.

	Model alone		Model + qcore9, hand identities				Model + qcore9, zero-config (qcore8)	
Model	Acc. %	Silent [CI]	Text layer	No page text	docTR	Tesseract	Text layer	No page text
SEC set 4 (new, held out): 72 statements, 5,351 numbers								
Haiku	96.86	168 [108, 246]	9 [0, 26] / 1.9	67 / 7.8	$52^{\dagger 1}$ / 12.0	$39^{\dagger 1}$ / 17.4	16 [3, 36]$^{\dagger 3}$ / 2.5 (19)	120 / 4.3 (125)
Opus	99.53	25 [7, 49]	0 [0, 0] / 1.0	17 / 0.7	11 / 7.0	14 / 10.1	3 [0, 9]$^{\dagger 3}$ / 1.2 (3)	20 / 1.1 (20)
GPT	79.63	1,090 [874, 1,324]	15 [2, 34]$^{\dagger 2}$ / 9.6	585 / 24.1	343 / 28.6	408 / 35.3	37 [17, 63]$^{\dagger 3}$ / 14.9 (51)	927 / 9.7 (942)
SEC sets 1–4 pooled, hand identities: 227 statements, 17,125 numbers								
Haiku, all qcore6	95.94	695 [536, 867]	32 [14, 55] / 2.5	352 / 7.2	$237^{\dagger 3}$ / 13.5	$211^{\dagger 3}$ / 18.2	–	–
qcore9 on set 4			28 [10, 50] / 2.5	352 / 7.2	$236^{\dagger 3}$ / 13.5	$211^{\dagger 3}$ / 18.2	–	–
Opus	98.98	174 [102, 258]	5 [0, 12] / 1.2	111 / 1.4	$80^{\dagger 1}$ / 8.5	76 / 11.7	–	–
GPT, all qcore6	76.20	4,076 [3,536, 4,629]	126 [82, 176]$^{\dagger 3}$ / 10.6	2,196 / 25.5	1,425$^{\dagger 2}$ / 31.0	1,530$^{\dagger 2}$ / 37.8	–	–
qcore9 on set 4			115 [71, 166]$^{\dagger 4}$ / 10.6	2,196 / 25.5	1,424$^{\dagger 2}$ / 31.0	1,530$^{\dagger 2}$ / 37.8	–	–

all hold on the truth; 9 failing linkbase relations excluded before writing). The bundles and page images were locked at 15:43:59Z (manifests `0b37a250...`, 76 files; `ba82b969...`, 152 files), the Haiku and Opus extractions at 15:54Z (`6ae22d31...`, 263 files) and GPT's at 16:12Z (`7156d227...`, 72 files), all before the freeze; the four manifests and the three package hashes were verified before the runs, after them and at report time. 90 runs (3 versions × 3 models × hand identities in four modes and three discovered settings in two), none crashed, none repeated, in three batches because OCR was not ready (text and no text 72 runs, Tesseract 9, docTR 9). OCR used qcore9's `ocr` command, whose file is identical to qcore6's, on page images pixel-identical to those the models read. The extractions were made as for set 3, by subagents, 12 batches of six statements per model, with set 3's prompt; Haiku left 2 blanks, Opus 0, GPT 10. The prompt does not forbid using printed totals to settle an unclear digit; of the per-statement notes (Haiku 67, Opus 52), 40 and 38 mention footing. The Opus re-read tier, run on set 3 as an add-on row, was not run. Scoring: legacy (primary, as sets 1–3) and strict (half a printed unit). Runtimes in the run records are inflated: qcore9's development grid shared the machine. Sources: `out/qcore9_heldout/SUMMARY.md`, `results.md`, `results.json`, `notes.json`; `sec_eval4/README.md`; `qcore9/CHANGES.md`; generator of these tables: `paper/gen_v9/gen_q9_tex.py`. *Refreeze.* At 19:59:26Z, after this run, qcore9 was refrozen (`c4401b943fcb8eb1...`; `qcore9/frozen_manifest.json` and `out/qcore9_work/FROZEN.md` now name that hash) because the first freeze's development grid broke one SEC development number in discovered-identity runs; `auto._one_slip` now needs a digit change of two units or more. The record says both freezes preceded any held-out run; the run records above (package checks 18:40:54Z, 19:19:23Z and 19:23:34Z, all `3ca510d3`) show the second came after this run. The change is in discovery only, so the hand-identity rows would be unchanged; the discovered and zero-config rows are those of the first freeze.

Table E.2: Paired differences in silent errors, qcore9 on SEC set 4 (legacy), sums over statements, 95% bootstrap intervals
vs alone: against the same model alone; *vs Opus alone* (Haiku and GPT): against Opus alone (25 silent errors); *vs qcore8, vs qcore6*: the same model, mode and identity setting; *vs hand*: discovered against hand identities, qcore9. 10,000 resamples, seed 0.

Model	Mode	Identities	Silent	vs alone	vs Opus alone	vs qcore8	vs qcore6	vs hand
Haiku	text	hand	9	−159 [−237, −100]	−16 [−43, +10]	−4 [−8, −1]	−4 [−8, −1]	–
Haiku	text	discovered	13	−155 [−234, −95]	−12 [−40, +15]	−3 [−7, 0]	−3 [−9, +3]	+4 [0, +11]
Haiku	text	hand + disc.	9	−159 [−237, −100]	−16 [−43, +10]	−4 [−8, −1]	−4 [−8, −1]	0 [0, 0]
Haiku	text	zero-config	16	−152 [−229, −94]	−9 [−36, +17]	−3 [−7, 0]	−15 [−44, +3]	+7 [+1, +15]
Haiku	no text	hand	67	−101 [−166, −52]	+42 [+16, +68]	0 [0, 0]	0 [0, 0]	–
Haiku	no text	discovered	120	−48 [−65, −33]	+95 [+52, +148]	−4 [−9, 0]	−5 [−11, −1]	+53 [+9, +112]
Haiku	no text	hand + disc.	56	−112 [−177, −63]	+31 [+6, +56]	0 [0, 0]	0 [0, 0]	−11 [−21, −3]
Haiku	no text	zero-config	120	−48 [−65, −33]	+95 [+52, +148]	−5 [−11, −1]	−4 [−9, 0]	+53 [+9, +112]
Haiku	OCR docTR	hand	52	−116 [−183, −66]	+27 [+2, +53]	−1 [−3, 0]	−1 [−3, 0]	–
Haiku	OCR Tesseract	hand	39	−129 [−196, −77]	+14 [−7, +34]	0 [0, 0]	0 [0, 0]	–
Opus	text	hand	0	−25 [−49, −7]	–	0 [0, 0]	0 [0, 0]	–
Opus	text	discovered	3	−22 [−46, −3]	–	0 [0, 0]	+1 [−3, +6]	+3 [0, +9]
Opus	text	hand + disc.	0	−25 [−49, −7]	–	0 [0, 0]	0 [0, 0]	0 [0, 0]
Opus	text	zero-config	3	−22 [−46, −3]	–	0 [0, 0]	+2 [−3, +9]	+3 [0, +9]
Opus	no text	hand	17	−8 [−22, 0]	–	0 [0, 0]	0 [0, 0]	–
Opus	no text	discovered	20	−5 [−12, 0]	–	0 [0, 0]	0 [0, 0]	+3 [0, +9]
Opus	no text	hand + disc.	17	−8 [−22, 0]	–	0 [0, 0]	0 [0, 0]	0 [0, 0]
Opus	no text	zero-config	20	−5 [−12, 0]	–	0 [0, 0]	0 [0, 0]	+3 [0, +9]
Opus	OCR docTR	hand	11	−14 [−29, −3]	–	0 [0, 0]	0 [0, 0]	–
Opus	OCR Tesseract	hand	14	−11 [−26, −2]	–	0 [0, 0]	0 [0, 0]	–
GPT	text	hand	15	−1,075 [−1,303, −863]	−10 [−38, +17]	−11 [−23, 0]	−11 [−23, 0]	–
GPT	text	discovered	27	−1,063 [−1,291, −853]	+2 [−28, +30]	−11 [−20, −4]	−15 [−25, −7]	+12 [+3, +23]
GPT	text	hand + disc.	15	−1,075 [−1,302, −864]	−10 [−38, +17]	−9 [−20, +1]	−10 [−21, 0]	0 [−4, +4]
GPT	text	zero-config	37	−1,053 [−1,278, −845]	+12 [−18, +43]	−14 [−25, −5]	−76 [−138, −33]	+22 [+10, +35]
GPT	no text	hand	585	−505 [−667, −362]	+560 [+443, +693]	0 [0, 0]	0 [0, 0]	–
GPT	no text	discovered	927	−163 [−202, −127]	+902 [+705, +1,117]	−19 [−37, −6]	−32 [−53, −15]	+342 [+206, +497]
GPT	no text	hand + disc.	554	−536 [−698, −391]	+529 [+415, +657]	−7 [−17, −1]	−8 [−21, +5]	−31 [−48, −13]
GPT	no text	zero-config	927	−163 [−202, −127]	+902 [+705, +1,117]	−15 [−25, −6]	−28 [−45, −14]	+342 [+206, +497]
GPT	OCR docTR	hand	343	−747 [−929, −582]	+318 [+241, +406]	−1 [−3, 0]	−1 [−3, 0]	–
GPT	OCR Tesseract	hand	408	−682 [−854, −526]	+383 [+290, +489]	0 [0, 0]	0 [0, 0]	–

Table E.4: SEC sets 1–4 pooled, hand identities (legacy)
17,125 numbers, 227 statements (sets 1–3: 11,774 numbers, frozen qcore6 held-out runs re-scored; they reproduce `out/qcore6_heldout/results.json` in 137 of 137 rows). *all qcore6*: qcore6 on set 4 too (comparable with the pooled rows of earlier versions); *qcore9 on set 4*: qcore6 on sets 1–3 and qcore9 on set 4. Differences against Opus alone are paired over statements.

Model	Mode	Set 4	Silent [CI]	To a person	vs Opus alone
Haiku	alone		695 [536, 867]	0.0%	
Haiku	text	all qcore6	32 [14, 55]	2.5%	−142 [−228, −68]
Haiku	text	qcore9	28 [10, 50]	2.5%	−146 [−231, −72]
Haiku	no text	all qcore6	352 [276, 438]	7.2%	+178 [+109, +251]
Haiku	no text	qcore9	352 [276, 438]	7.2%	+178 [+109, +251]
Haiku	OCR docTR	all qcore6	237 [183, 295]$^{\dagger 3}$	13.5%	+63 [−5, +125]
Haiku	OCR docTR	qcore9	236 [182, 294]$^{\dagger 3}$	13.5%	+62 [−5, +124]
Haiku	OCR Tesseract	all qcore6	211 [161, 266]$^{\dagger 3}$	18.2%	+37 [−32, +100]
Haiku	OCR Tesseract	qcore9	211 [161, 266]$^{\dagger 3}$	18.2%	+37 [−32, +100]
Opus	alone		174 [102, 258]	0.0%	
Opus	text	all qcore6	5 [0, 12]	1.2%	
Opus	text	qcore9	5 [0, 12]	1.2%	
Opus	no text	all qcore6	111 [64, 166]	1.4%	
Opus	no text	qcore9	111 [64, 166]	1.4%	
Opus	OCR docTR	all qcore6	80 [44, 123]$^{\dagger 1}$	8.5%	
Opus	OCR docTR	qcore9	80 [44, 123]$^{\dagger 1}$	8.5%	
Opus	OCR Tesseract	all qcore6	76 [43, 117]	11.7%	
Opus	OCR Tesseract	qcore9	76 [43, 117]	11.7%	
GPT	alone		4,076 [3,536, 4,629]	0.0%	
GPT	text	all qcore6	126 [82, 176]$^{\dagger 3}$	10.6%	−48 [−143, +38]
GPT	text	qcore9	115 [71, 166]$^{\dagger 4}$	10.6%	−59 [−153, +28]
GPT	no text	all qcore6	2,196 [1,921, 2,490]	25.5%	+2,022 [+1,769, +2,290]
GPT	no text	qcore9	2,196 [1,921, 2,490]	25.5%	+2,022 [+1,769, +2,290]
GPT	OCR docTR	all qcore6	1,425 [1,224, 1,639]$^{\dagger 2}$	31.0%	+1,251 [+1,069, +1,449]
GPT	OCR docTR	qcore9	1,424 [1,223, 1,638]$^{\dagger 2}$	31.0%	+1,250 [+1,068, +1,448]
GPT	OCR Tesseract	all qcore6	1,530 [1,331, 1,745]$^{\dagger 2}$	37.8%	+1,356 [+1,166, +1,559]
GPT	OCR Tesseract	qcore9	1,530 [1,331, 1,745]$^{\dagger 2}$	37.8%	+1,356 [+1,166, +1,559]

Table E.5: Every correct number broken on SEC set 4 (legacy; strict breaks the same numbers), with its cause

Each cell once, with every run that broke it (^{f}: flagged in that run; all others silent). No no-text run broke a number. Causes: (1) WAT's PDF text layer prints the closing parenthesis as a separate word, so a negative number reads as positive (a tokenisation gap in every version); (2) a GPT row shift that qcore9's change 3 partly repaired and identity completion then finished on a correct cell; (3) OCR look-alike repairs, the class of set 3's OCR breaks; (4) a qcore6 discovery break fixed in qcore7.

Cell	Truth	Changed to	Cause	Runs that broke it
WAT cash flow: Net cash used in financing activities (2025)	-174,000,000	174,000,000	(1)	Haiku q9 disc., text; Haiku q9 ZC, text; Haiku q8 disc., text; Haiku q8 ZC, text; Haiku q6 disc., text^f; Opus q9 disc., text; Opus q9 ZC, text; Opus q8 disc., text; Opus q8 ZC, text; Opus q6 disc., text^f; GPT q9 disc., text; GPT q9 ZC, text; GPT q8 disc., text; GPT q8 ZC, text; GPT q6 disc., text
WAT cash flow: Net cash used in financing activities (2026)	-337,000,000	337,000,000	(1)	Haiku q9 disc., text; Haiku q9 ZC, text; Haiku q8 disc., text; Haiku q8 ZC, text; Haiku q6 disc., text; Opus q9 disc., text; Opus q9 ZC, text; Opus q8 disc., text; Opus q8 ZC, text; Opus q6 disc., text; GPT q9 disc., text; GPT q9 ZC, text; GPT q8 disc., text; GPT q8 ZC, text; GPT q6 disc., text
WAT cash flow: Net cash provided by (used in) investing activities (2025)	-85,000,000	85,000,000	(1)	Haiku q9 disc., text; Haiku q9 ZC, text; Haiku q8 disc., text; Haiku q8 ZC, text; Haiku q6 disc., text^f; Opus q9 disc., text; Opus q9 ZC, text; Opus q8 disc., text; Opus q8 ZC, text; Opus q6 disc., text^f; GPT q9 disc., text; GPT q9 ZC, text; GPT q8 disc., text; GPT q8 ZC, text; GPT q6 disc., text
WAT income statement: Net (loss) income per basic common share (2026)	-2.31	2.31	(1)	GPT q9, text; GPT q8, text; GPT q6, text; GPT q6 disc., text; GPT q6 hand+disc., text
WST cash flow: Decrease (increase) in other current assets (2024)	-5,800,000	5,700,000	(2)	GPT q9, text; GPT q9 hand+disc., text
DPZ balance sheet: Advertising fund assets, restricted (2024)	103,396,000	103,316,000	(3)	Haiku q9, docTR; Haiku q8, docTR; Haiku q6, docTR
GRMN balance sheet: Deferred income tax assets (2024)	822,521,000	822,623,000	(3)	Haiku q9, Tess.; Haiku q8, Tess.; Haiku q6, Tess.
ALB balance sheet: Common stock, $.01 par value, . . . (2025)	1,178,000	2,025,000	(4)	Haiku q6 disc., text; Opus q6 disc., text; GPT q6 disc., text

Every run of qcore9, qcore8 and qcore6 on set 4 and the run-by-run hand-identity check are in Supplement S5.

F Development grid of frozen qcore9

The grid of the version run on set 4. This is the development grid of qcore9's first freeze (`3ca510d3`), the version run on set 4, copied before the package was refrozen (`/home/claude/q9/logs/dev_report9_freeze1.json`, `.md`; written by `out/qcore9_work/dev_report9.py`). Its one new break, CAT's six-month net income to common holders (6,142 to 6,141 M) in discovered and hand + discovered text runs of the SEC development filings, led to the refreeze (Section 4). Set names are those of the development scripts: A, B, C = SEC sets 1–3; D = CORD test; E = notes tables; F = real scans; G = synthetic 02–05; H, I = qcore7's and qcore8's fresh sets. **Development data only.** Every set below was spent before qcore9 was written, and qcore9 was tuned on them, set I (qcore8's fresh set) included. Frozen qcore9 was run on every (set, model, extraction, mode, identity setting) of the grid; the qcore8 runs are qcore8's own frozen grid and, for set I, its held-out run. Hand-identity check, run by run: 352 hand runs, 350 with qcore8's fixes, 330 with its tier 2 items (22 differ, all through change 3). Missing runs: 0.

Table F.2: Development runs in which frozen qcore9 broke a correct number

Per run: numbers broken by qcore9, by qcore8 in the same run, and how many of qcore9's were also broken by qcore8.

Set	Extraction	Mode	Identities	Scoring	qcore9	qcore8	also qcore8
SEC set 1	luna	ocr-doctr	hand	legacy	2	2	2
SEC set 2	haiku	ocr-tesseract	hand	legacy	1	1	1
SEC set 2	opus	ocr-doctr	hand	legacy	1	1	1
SEC set 2	luna	ocr-tesseract	hand	legacy	1	1	1
SEC set 3	haiku	ocr-doctr	hand	legacy	2	2	2
SEC set 3	haiku	ocr-tesseract	hand	legacy	1	1	1
SEC set 3	luna	text	hand	legacy	2	2	2
SEC set 3	luna	ocr-tesseract	hand	legacy	1	1	1
Notes tables	haiku	text	hand	legacy	1	1	1
Notes tables	opus	text	hand	legacy	1	1	1
Notes tables	luna	text	hand	legacy	1	1	1
SEC dev	haiku poor	text	discovered	legacy	1	0	0
SEC dev	haiku poor	text	hand + disc.	legacy	1	0	0
SEC dev	opus poor	text	discovered	legacy	1	0	0
SEC dev	opus poor	text	hand + disc.	legacy	1	0	0
SEC dev	opus	text	discovered	legacy	1	0	0
SEC dev	opus	text	hand + disc.	legacy	1	0	0

The full development grid, per set and identity setting, is in Supplement S6.

The refrozen package's grid. qcore9 as refrozen (`c4401b94`, not run on set 4) was run on the same 1,330-run grid (`out/qcore9_work/dev_summary9.md`, `dev_report9.md`, `dev_report9.json`): no crashed or missing run; no number broken with discovered identities, hand + discovered or zero-config on any set (qcore8: 20 on set I); 14 hand-mode breaks in 11 runs, each

Table F.1: Development data, not held out: frozen qcore8 → qcore9's first freeze (the version run on set 4) on every spent set, pooled over models, extractions and modes; legacy scoring, strict for sets H and I. qcore9 was tuned on these sets. Hand identities: 350 of 352 runs return qcore8's fixes and 330 its tier 2 items. Every row: Appendix F.

Set	Ids	Silent	Broken	Fixed	Flag. %
SEC set 1	hand	885 → 885	2 → 2	563 → 563	14.3 → 14.4
SEC set 1	ZC	593 → 575	0 → 0	259 → 284	6.0 → 6.3
SEC set 2	hand	1,615 → 1,614	3 → 3	1,001 → 1,001	12.9 → 12.9
SEC set 2	ZC	1,072 → 1,054	0 → 0	381 → 406	6.1 → 6.3
SEC set 3	hand	2,304 → 2,241	6 → 6	1,108 → 1,108	16.2 → 16.3
SEC set 3	ZC	1,525 → 1,460	0 → 0	449 → 473	8.0 → 8.3
CORD test	hand	365 → 365	0 → 0	28 → 28	20.9 → 20.9
CORD test	ZC	144 → 144	0 → 0	25 → 25	21.5 → 22.2
Notes tables	hand	54 → 54	3 → 3	8 → 8	7.1 → 7.1
Notes tables	ZC	95 → 96	0 → 0	0 → 1	8.6 → 8.7
Real scans	hand	13 → 13	0 → 0	0 → 0	16.2 → 16.2
Real scans	ZC	5 → 5	0 → 0	0 → 0	20.6 → 20.6
Synthetic 02–05	hand	4 → 4	0 → 0	0 → 0	2.1 → 2.1
Synthetic 02–05	ZC	4 → 4	0 → 0	0 → 0	2.9 → 2.9
Set H (qcore7 fresh)	hand	38 → 37	0 → 0	253 → 253	8.5 → 8.5
Set H (qcore7 fresh)	ZC	110 → 103	0 → 0	138 → 167	8.5 → 8.5
Set I (qcore8 fresh)	hand	91 → 90	1 → 0	622 → 622	7.8 → 7.8
Set I (qcore8 fresh)	ZC	404 → 375	16 → 0	367 → 435	9.6 → 9.3
SEC dev	hand	130 → 130	0 → 0	154 → 154	7.7 → 7.8
SEC dev	ZC	181 → 169	0 → 0	89 → 118	6.5 → 6.5
Notes dev	hand	103 → 103	0 → 0	34 → 34	5.3 → 5.3
Notes dev	ZC	96 → 92	0 → 0	11 → 14	6.6 → 6.6
CORD train	hand	33 → 33	0 → 0	63 → 63	2.9 → 2.9
CORD train	ZC	73 → 73	0 → 0	31 → 31	4.4 → 4.4
CORD val.	hand	64 → 64	0 → 0	36 → 36	10.4 → 10.4
CORD val.	ZC	115 → 115	0 → 0	9 → 9	10.3 → 10.3
Real-scan dev	hand	8 → 8	0 → 0	0 → 0	24.9 → 24.9
Real-scan dev	ZC	16 → 16	0 → 0	0 → 0	18.6 → 18.6
Synthetic 00–01	hand	22 → 22	0 → 0	354 → 354	4.4 → 4.4
Synthetic 00–01	ZC	166 → 154	0 → 0	163 → 200	5.6 → 5.5

a qcore8 break on the same cell with the same value; the same hand-mode check (352 runs, 350 with qcore8's fixes, 330 with its tier 2 items). Zero-config silent errors, pooled: SEC set 1 593 to 575, set 2 1,072 to 1,054, set 3 1,525 to 1,461, notes tables 95 to 96. On the SEC development sets (sets 1–3 and the development filings), zero-config against hand: with the text layer 166 against 79 (qcore8: 216 against 126), without 3,013 against 2,061 (qcore8: 3,075 against 2,061). Its summary describes both freezes as preceding any held-out run; the set-4 run (18:40–19:19Z) preceded the second.

G Frozen qcore8 on the second fresh held-out set

Frozen qcore8 (29 files, combined sha256 `405186160d8d9a9ef0f1b846630e18ad1dfae94871fdb42d120fd627cc616558`, frozen 2026-10-16T21:54:50Z, 71 tests), frozen qcore7 and frozen qcore6 were run once, 2026-10-16 22:32:57–23:05:26Z, on 18 bundles: 12 from four generators written for this test (`gen_domains8/`: budget, census, inventory, utilbank; 103 documents, 9,860 numbers) and 6 from the earlier generators with new seeds 09–10 and other layout variants (96 documents, 4,008 numbers). The data were locked by a hash manifest (387 files, combined sha256 `f1008f45a959cc58...`) at 19:43:36Z, before any qcore8 change; the three package hashes and the data hash were checked at the start and at the end. 1,296 runs (3 versions × 18 bundles × 3 extractions × 2 modes × 4 identity settings), none crashed. The extractions are the true values (*clean*), and the true values with injected errors (about 1.5% and 5%: `inject8.py`, qcore7's injector plus a dropped decimal point); under strict scoring 210 and 691 numbers are wrong, under the legacy rule 191 and 640. qcore8 was frozen three times before this run (`d2ce394b`, `7aac7220`, then `40518616`); each earlier freeze's development grid showed a new zero-config break (one notes-table number per model; two SEC set 3 numbers, a diluted share count read from the diluted earnings-per-share line), fixed by a guard in the inferred schema's label rule. Neither the held-out documents, their truth nor their extractions were opened before the run. Sources: `out/qcore8_heldout/SUMMARY.md`, `results.md`, `results.json`; `qcore8/CHANGES.md`; generator of these tables: `paper/gen_v8/gen_q8_tex.py`.

Table G.1: Frozen qcore8 against frozen qcore7 on the second fresh held-out set, one run: 18 synthetic bundles, 199 documents, 13,868 numbers. *NEW*: 12 bundles of four document families written for this test and never seen in development (budget-vs-actual reports, hospital and school census, inventory counts, utility bills with bank statements; 9,860 numbers); *OLD*: 6 bundles of the earlier generators with new seeds and layout variants (4,008 numbers). The extractions are the true values with injected errors; no model read these pages. Strict scoring. Cells: silent errors (ZC qcore8: with its 95% bootstrap interval over documents) / % of numbers to a person; $^{\dagger n}$: *n* correct numbers broken. *Hand*: hand identities and schema, identical in qcore6, qcore7 and qcore8 in all 108 runs. *ZC*: zero-config. Paired differences over the same resamples. Identity coverage (share of numbers in some identity): hand 94%, ZC qcore7 50–70%, ZC qcore8 75–83%. Every row: Appendix G.

				Zero-config		ZC qcore8, paired silent diff.		NEW	OLD
Extraction	Mode	Alone	Hand	qcore7	qcore8	− qcore7	− hand	silent: hand / ZC qcore7 / ZC qcore8	
clean	text	0	0 / 2.4	$7^{\dagger 7}$ / 3.3	2 [0, 5]$^{\dagger 2}$ / 7.0	−5 [−13, 0]	+2 [0, +5]	0 / $7^{\dagger 7}$ / $2^{\dagger 2}$	0 / 0 / 0
	no text		0 / 1.2	0 / 0.6	0 [0, 0] / 5.6	0 [0, 0]	0 [0, 0]	0 / 0 / 0	0 / 0 / 0
low, 1.5%	text	210	12 / 4.9	$51^{\dagger 11}$ / 3.5	27 [16, 39]$^{\dagger 2}$ / 7.6	−24 [−38, −13]	+15 [+2, +28]	10 / $36^{\dagger 11}$ / $18^{\dagger 2}$	2 / 15 / 9
	no text		22 / 7.8	87 / 4.2	58 [42, 76] / 8.5	−29 [−42, −17]	+36 [+20, +52]	20 / 65 / 46	2 / 22 / 12
high, 5%	text	691	$15^{\dagger 1}$ / 10.4	$182^{\dagger 13}$ / 5.4	106 [73, 145]$^{\dagger 12}$ / 12.9	−76 [−107, −50]	+91 [+59, +130]	11 / $141^{\dagger 13}$ / $83^{\dagger 12}$	4 / 41 / 23
	no text		42 / 20.2	413 / 6.4	211 [168, 257] / 16.1	−202 [−256, −155]	+169 [+128, +215]	34 / 326 / 168	8 / 87 / 43

Table G.2: Every correct number broken on the second fresh held-out set (strict)

Every break is silent and in text mode; no no-text run broke a number. Cells are named document | field | row; *reason* is the scorer's class of the change.

Extraction	Pipeline	Broken	Cells (truth → changed to)	Reason
clean	qcore6 zero-config	9	utilbank_10 | 2025-06_bank_statement | Check 1270: check_amount 673.05→-673.05; utilbank_10 | 2025-06_bank_statement | Check 1271: check_amount 557.85→-557.85; utilbank_10 | 2025-06_bank_statement | Total checks: checks_total 1230.9→-1230.9; utilbank_10 | 2025-07_bank_statement | Check 1272: check_amount 464.23→-464.23; utilbank_10 | 2025-07_bank_statement | Check 1273: check_amount 429.82→-429.82; utilbank_10 | 2025-07_bank_statement | Check 1274: check_amount 708.11→-708.11; utilbank_10 | 2025-07_bank_statement | Total checks: checks_total 1602.16→-1602.16; utilbank_10 | 2025-08_bank_statement | Check 1275: check_amount 542.03→-542.03; utilbank_10 | 2025-09_bank_statement | Check 1276: check_amount 855.58→-855.58	7 wrong sign, 2 misread
clean	qcore7 zero-config	7	utilbank_10 | 2025-06_bank_statement | Check 1270: check_amount 673.05→-673.05; utilbank_10 | 2025-06_bank_statement | Check 1271: check_amount 557.85→-557.85; utilbank_10 | 2025-07_bank_statement | Check 1272: check_amount 464.23→-464.23; utilbank_10 | 2025-07_bank_statement | Check 1273: check_amount 429.82→-429.82; utilbank_10 | 2025-07_bank_statement | Check 1274: check_amount 708.11→-708.11; utilbank_10 | 2025-08_bank_statement | Check 1275: check_amount 542.03→-542.03; utilbank_10 | 2025-09_bank_statement | Check 1276: check_amount 855.58→-855.58	7 wrong sign
clean	qcore8 zero-config	2	utilbank_10 | 2025-08_bank_statement | Check 1275: check_amount 542.03→-542.03; utilbank_10 | 2025-09_bank_statement | Check 1276: check_amount 855.58→-855.58	2 wrong sign
low, 1.5%	qcore6 discovered	1	census_10 | wk02_20260914 | Sun 20 Sep: admissions 13→11	1 misread
low, 1.5%	qcore6 zero-config	9	utilbank_10 | 2025-06_bank_statement | Check 1270: check_amount 673.05→-673.05; utilbank_10 | 2025-06_bank_statement | Check 1271: check_amount 557.85→-557.85; utilbank_10 | 2025-06_bank_statement | Total checks: checks_total 1230.9→-1230.9; utilbank_10 | 2025-07_bank_statement | Check 1272: check_amount 464.23→-464.23; utilbank_10 | 2025-07_bank_statement | Check 1273: check_amount 429.82→-429.82; utilbank_10 | 2025-07_bank_statement | Check 1274: check_amount 708.11→-708.11; utilbank_10 | 2025-07_bank_statement | Total checks: checks_total 1602.16→-1602.16; utilbank_10 | 2025-08_bank_statement | Check 1275: check_amount 542.03→-542.03; utilbank_10 | 2025-09_bank_statement | Check 1276: check_amount 855.58→-855.58	7 wrong sign, 2 misread
low, 1.5%	qcore7 discovered	1	census_10 | wk02_20260914 | Sun 20 Sep: admissions 13→11	1 misread
low, 1.5%	qcore7 zero-config	11	inventory_09 | 2026-01_count_sheet | ANA-128: open_qty 130→128, received_qty 48→0, sold_qty 61→41, book_qty 117→87; utilbank_10 | 2025-06_bank_statement | Check 1270: check_amount 673.05→-673.05; utilbank_10 | 2025-06_bank_statement | Check 1271: check_amount 557.85→-557.85; utilbank_10 | 2025-07_bank_statement | Check 1272: check_amount 464.23→-464.23; utilbank_10 | 2025-07_bank_statement | Check 1273: check_amount 429.82→-429.82; utilbank_10 | 2025-07_bank_statement | Check 1274: check_amount 708.11→-708.11; utilbank_10 | 2025-08_bank_statement | Check 1275: check_amount 542.03→-542.03; utilbank_10 | 2025-09_bank_statement | Check 1276: check_amount 855.58→-855.58	4 misread, 7 wrong sign
low, 1.5%	qcore8 discovered	2	budget_11 | m05_february_2026 | Manufacturing overhead: ytd_actual 368.8→378.8; census_10 | wk02_20260914 | Sun 20 Sep: admissions 13→11	2 misread
low, 1.5%	qcore8 zero-config	2	utilbank_10 | 2025-08_bank_statement | Check 1275: check_amount 542.03→-542.03; utilbank_10 | 2025-09_bank_statement | Check 1276: check_amount 855.58→-855.58	2 wrong sign

Extraction	Pipeline	Broken	Cells (truth → changed to)	Reason
high, 5%	qcore6 hand	1	budget_11 | m05_february_2026 | General and administrative: remaining_budget 326→331	1 misread
high, 5%	qcore6 discovered	4	budget_09 | m05_november_2025 | Contracted services: month_variance 2023→2973; budget_09 | m05_november_2025 | Equipment maintenance: month_variance -251→-234; budget_09 | m06_december_2025 | Other operating: month_variance 318→385; budget_11 | m06_march_2026 | Product sales: ytd_variance 187.1→177.2	4 misread
high, 5%	qcore6 hand + disc.	1	budget_11 | m05_february_2026 | General and administrative: remaining_budget 326→331	1 misread
high, 5%	qcore6 zero-config	12	inventory_09 | 2026-01_count_sheet | ANA-128: counted_qty 119→87, unit_cost 6.7→2.91, extended_value 797.3→253.17; utilbank_10 | 2025-06_bank_statement | Check 1270: check_amount 673.05→-673.05; utilbank_10 | 2025-06_bank_statement | Check 1271: check_amount 557.85→-557.85; utilbank_10 | 2025-06_bank_statement | Total checks: checks_total 1230.9→-1230.9; utilbank_10 | 2025-07_bank_statement | Check 1272: check_amount 464.23→-464.23; utilbank_10 | 2025-07_bank_statement | Check 1273: check_amount 429.82→-429.82; utilbank_10 | 2025-07_bank_statement | Check 1274: check_amount 708.11→-708.11; utilbank_10 | 2025-07_bank_statement | Total checks: checks_total 1602.16→-1602.16; utilbank_10 | 2025-08_bank_statement | Check 1275: check_amount 542.03→-542.03; utilbank_10 | 2025-09_bank_statement | Check 1276: check_amount 855.58→-855.58	5 misread, 7 wrong sign
high, 5%	qcore7 hand	1	budget_11 | m05_february_2026 | General and administrative: remaining_budget 326→331	1 misread
high, 5%	qcore7 discovered	3	budget_09 | m05_november_2025 | Contracted services: month_variance 2023→2973; budget_09 | m05_november_2025 | Equipment maintenance: month_variance -251→-234; budget_09 | m06_december_2025 | Other operating: month_variance 318→385	3 misread
high, 5%	qcore7 hand + disc.	1	budget_11 | m05_february_2026 | General and administrative: remaining_budget 326→331	1 misread
high, 5%	qcore7 zero-config	13	inventory_09 | 2026-01_count_sheet | ANA-128: received_qty 48→0, sold_qty 61→41, book_qty 117→87, counted_qty 119→87, unit_cost 6.7→2.91, extended_value 797.3→253.17; utilbank_10 | 2025-06_bank_statement | Check 1270: check_amount 673.05→-673.05; utilbank_10 | 2025-06_bank_statement | Check 1271: check_amount 557.85→-557.85; utilbank_10 | 2025-07_bank_statement | Check 1272: check_amount 464.23→-464.23; utilbank_10 | 2025-07_bank_statement | Check 1273: check_amount 429.82→-429.82; utilbank_10 | 2025-07_bank_statement | Check 1274: check_amount 708.11→-708.11; utilbank_10 | 2025-08_bank_statement | Check 1275: check_amount 542.03→-542.03; utilbank_10 | 2025-09_bank_statement | Check 1276: check_amount 855.58→-855.58	6 misread, 7 wrong sign
high, 5%	qcore8 hand	1	budget_11 | m05_february_2026 | General and administrative: remaining_budget 326→331	1 misread
high, 5%	qcore8 discovered	1	budget_11 | m05_february_2026 | Manufacturing overhead: ytd_actual 368.8→378.8	1 misread
high, 5%	qcore8 hand + disc.	1	budget_11 | m05_february_2026 | General and administrative: remaining_budget 326→331	1 misread
high, 5%	qcore8 zero-config	12	inventory_09 | 2026-01_count_sheet | ANA-128: received_qty 48→0, sold_qty 61→41, book_qty 117→87, counted_qty 119→87, variance_qty 2→0, unit_cost 6.7→2.91, extended_value 797.3→253.17; utilbank_10 | 2025-07_bank_statement | Check 1272: check_amount 464.23→-464.23; utilbank_10 | 2025-07_bank_statement | Check 1273: check_amount 429.82→-429.82; utilbank_10 | 2025-07_bank_statement | Check 1274: check_amount 708.11→-708.11; utilbank_10 | 2025-08_bank_statement | Check 1275: check_amount 542.03→-542.03; utilbank_10 | 2025-09_bank_statement | Check 1276: check_amount 855.58→-855.58	7 misread, 5 wrong sign

Every run, the NEW and OLD groups, paired differences, legacy scoring, injected errors by kind and by domain, discovered identities by source and qcore8's development grid are in Supplement S7.

H Discovered identities

Tables S2.14 and S2.16 (Supplement S2) give every held-out run with discovered identities. Table H.1 gives the precision of discovery on development data: how many discovered identities hold on the true values, and how many lie in the span of the hand identities of the same documents (an identity outside the span can still be true: a tie the hand file does not state).

Table H.1: Discovered identities on development data (selected rows of `out/qcore6_work/dev_summary6.md`)

Set	Extraction	Mode	Schema	Discovered	Hold on truth	In hand span / linear	Coverage hand / disc. / both
SEC dev	Haiku poor	text	hand	501	501	456 / 501	82.4 / 83.7 / 92.2%
SEC dev	Haiku poor	text	auto	485	484	444 / 485	82.4 / 77.8 / 91.1%
SEC dev	Opus poor	notext	hand	515	514	474 / 515	82.4 / 89.3 / 91.8%
CORD train	gold	text	auto	2,183	2,176	740 / 1,569	73.6 / 64.2 / 80.4%
Notes dev	Opus	text	auto	1,046	1,046	396 / 1,046	44.8 / 57.0 / 70.8%
Notes dev	Haiku	notext	hand	1,091	1,070	396 / 1,091	44.8 / 60.4 / 72.7%
Real-scan dev	Haiku	notext	hand	22	22	1 / 22	98.3 / 28.4 / 98.3%
Synthetic	corp_fin_00 Opus	text	auto	312	310	269 / 312	96.4 / 91.9 / 96.4%
Synthetic	invoices_00 Opus	text	auto	98	98	17 / 22	96.6 / 53.2 / 96.6%

Without page text, discovery rests on the extraction alone: on the notes development tables 21 of 1,091 discovered identities do not hold on the truth (1 of 1,059 with page text), and they added only flags (77 false flags on the error-free Opus run). On development data zero-config broke no number but found fewer fixes than the hand files (SEC text 45 against 73; CORD with injected errors 11–76 against 43–128).

I Further SEC results

Table I.1: OCR words fed to frozen qcore3 as if they were a text layer, against qcore4's two-reader mode, SEC held-out set 1
Silent / flagged / broken. The two runs used separate OCR pipelines of the same engines (source: `ocr_mode/results.md`, `out/qcore4_heldout/results.md`). With docTR the page scale was wrong for 1,244 of 2,022 dollar amounts, because "In millions" came back as "mullions".

OCR	Extractor	qcore3, as text layer	qcore4, two readers
docTR	Haiku	39 / 1,485 / 93	47 / 234 / 0
docTR	Opus	34 / 1,471 / 102	10 / 168 / 0
Tesseract	Haiku	13 / 2,009 / 32	38 / 392 / 0
Tesseract	Opus	13 / 2,009 / 32	9 / 283 / 0
none	Haiku	57 / 155 / 0	57 / 155 / 0
none	Opus	13 / 62 / 0	13 / 62 / 0
Opus alone, no checker		30 / 0 / 0	

Set 1's identities were first inferred from layout and fitted on the truth. We later obtained the XBRL calculation linkbases of 17 of its 20 filings; they give 752 identities, all of which hold on the truth, and confirm 621 of our 1,054. Rebuilding the held-out bundle with the filers' identities and re-running frozen qcore3 changed no fix and no silent count in text mode and cut flagged numbers by about 15%; in no-text mode one error per extractor moved from flagged to silent. Set 2 used the linkbases from the start.
The filers' linkbase identities against set 1's, and qcore3 against qcore4 on the development filings, are in Supplement S9.

J Changes made on development data

Every version was changed on development data only, frozen, and run once on held-out data (Section 4). Table J.1 condenses the history from the first private-equity checker to qcore9; the main text describes qcore6's discovery and re-read tier (Sections 3.5 and 3.6) and the changes of qcore7, qcore8 and qcore9. With hand identities, qcore6 returned qcore5's fixes and tier 2 items in every held-out run, and qcore7 and qcore8 returned qcore6's; qcore9's change 3 is the one exception (Section 5.1). The full record of each version, with what was tried and not kept, the development costs and the development tables of qcore4 and qcore5, is in Supplement S10; its sources are each package's `CHANGES.md` and the development summaries under `out/qcore*_work/`.

Table J.1: Every frozen version: what it was tuned on, what it changed and what it was run on once (freeze times in Appendix K)

Version	Frozen (Oct. 2026)	Tuned on	Main changes	Tested once on
PE v1	14th 05:28	real fund Fund R; synthetic funds 01–09	the first frozen checker, for private-fund packets	funds 11–19; 360 attack cases
PE v2	14th 19:19	the 360 attack cases; v1 regression runs	fixes for the five v1 defects (Supplement S1)	funds 31–39; 430 fresh attack cases
qcore	15th 03:40	none (domain-neutral port of PE v2)	no domain code: identities, schema and adapter as inputs	synthetic domains, zero-shot
qcore2	15th 04:54	synthetic-domain dev bundles 00–01	prose and column reads, label and heading rules, number formats	synthetic bundles 02–05; CORD, SEC zero-shot
qcore3	15th 06:44	CORD train, validation; 10 SEC dev filings	unit statements and headers, own-line confirmation of swaps and flips, CORD adapter (23 tests)	CORD test; SEC sets 1, 2
qcore4	15th 18:33	SEC dev filings: poor scans, OCR, injected errors; CORD train, val.	OCR mode with the two-reader rule, scale vote, OCR cleanup, product terms (31 tests)	SEC 1, 2; CORD; notes; real scans
qcore5	16th 03:28	+ notes tables of the 10 dev filers; synthetic dev; PE validation	the gate: rules (a) and (b), joint guard, no power-of-1000 change (39 tests)	the same, all modes, three models
qcore6	16th 13:14	+ real-scan dev split; real re-reads of dev crops	zero-config discovery and schema inference, re-read tier, a scan's text layer read as OCR (56 tests)	SEC 1, 2, 3 (new); CORD; notes; real scans; synthetic 02–05
qcore7	16th 16:40	+ every qcore6 held-out set (now spent)	zero-config only: label cells, independent backing, aggregate ties, weak labels (65 tests)	fresh synthetic 06–08 (set H)
qcore8	16th 21:54 (3rd freeze)	+ set H (now spent)	zero-config only: more templates, tolerances, three label guards (71 tests)	set I: 4 new families; 09–10
qcore9	18th 18:27 (1st freeze)	+ set I (now spent)	changes 1–5 of Section 3.5; change 3 in every page-reading mode (79 tests)	SEC 4 (new), with qcore8, qcore6

K Reproduction

The accompanying artifact contains the code of every frozen version with its manifest, the bundle builders for the synthetic domains, CORD and SEC, the degradation scripts, the adversarial generators and the scoring scripts, together with the extractions and result files behind every table and figure. Freeze times: PE v1 2026-10-14T05:28:12Z, PE v2 2026-10-14T19:19:22Z, qcore 2026-10-15T03:40:40Z, qcore2 2026-10-15T04:54:42Z, qcore3 2026-10-15T06:44:43Z, qcore4 2026-10-15T18:33:58Z (combined sha256 of its 23 files `599928a82f855fe7...`), qcore5 2026-10-16T03:28:46Z (24 files, `97351ded97dbd93f...`), qcore6 2026-10-16T13:14:47Z (28 files, `86debea05f95a6d2...`), qcore7 2026-10-16T16:40:49Z (28 files, `51d9a35236ca553c...`), qcore8 2026-10-16T21:54:50Z (29 files, `405186160d8d9a9e...`; earlier freezes `d2ce394b`, `7aac7220` superseded before any held-out run), qcore9 2026-10-18T18:27:04Z (30 files, `3ca510d3ea178550...`, freeze 1, the version run on set 4 and the one every qcore9 result describes; refrozen 19:59:26Z as freeze 2, `c4401b943fcb8eb1...`, which `qcore9/` now matches and which has not been tested on held-out data; the freeze-1 code that produced every set-4 result is `qcore9_f1/` (package name `qcore9`; copy it to a folder named `qcore9` to reproduce `3ca510d3...` and run its 79 tests; its `frozen_manifest.json` holds reconstructed per-file hashes, as the original freeze-1 manifest was overwritten at freeze 2), and its development runs are in `/home/claude/q9/runs/qcore9_freeze1` with the report `/home/claude/q9/logs/dev_report9_freeze1.md` (copied to `paper/gen_v9/`); recompute with `find qcore9 -path '*__pycache__*' -prune -o \( -name '*.py' -o -name '*.md' \) -print | sort | xargs sha256sum | sha256sum`; per-file hashes in `qcore9/frozen_manifest.json`). The qcore3 test suite has 23 tests, the qcore4 suite 31, the qcore5 suite 39, the qcore6 suite 56, the qcore7 suite 65, the qcore8 suite 71 and the qcore9 suite 79 (`python -m pytest qcore9/tests -q`). SEC set 4: the builder (`sec_eval4/build/build_bundle4.py`, `degrade4.py`, `check4.py`), the extraction prompt and batches (`sec_eval4/extract/`), the lock manifests, the held-out harness and report (`out/qcore9_heldout/heldout9.py`, `report9.py`, `summary9.py`, `ocr_D.sh`) with the start and end checks, the development-grid scripts (`out/qcore9_work/dev9.py`, `dev_report9.py`) and this paper's table generator (`paper/gen_v9/gen_q9_tex.py`) are included. The four new generators (`gen_domains8/`), the qcore8 held-out builder, injector, harness and report (`out/qcore8_heldout/build/build_heldout8.py`, `inject8.py`, `heldout8.py`, `results_md8.py`) with the locked data and the start and end hash checks, and the development-grid scripts (`out/qcore8_work/dev8.py`, `dev_report8.py`) are included. The qcore7 fresh held-out builder, error injector and harness (`out/qcore7_heldout/build_heldout7.py`, `inject7.py`, `heldout7.py`), with the locked data, and the development-grid scripts (`out/qcore7_work/dev7.py`, `dev_report7.py`) are included. The SEC set 3 builder, the re-read request maker and the held-out harness of qcore6 (`out/qcore6_heldout/heldout_qcore6.py`) are included. The SEC set 2 builder, the notes-table and real-scan builders, the calculation-linkbase reader, the baseline script and the OCR scripts are included.